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# Training-Free Cultural Alignment of Large Language Models via Persona Disagreement

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## Abstract

1 Large language models are increasingly deployed in decisions that turn on moral  
2 judgement, from content moderation to clinical prioritisation, yet they answer as  
3 if the whole world thinks like the West. The Moral Machine experiment [Awad  
4 et al., 2018] showed this is wrong at scale: 40 million judgments across 233  
5 countries reveal that moral preferences are systematically structured by culture, and  
6 a model that ignores this variation does not merely underperform. It imposes one  
7 society’s intuitions on all others. Existing fixes do not scale to global deployment:  
8 fine-tuning needs per-country preference data and GPU budgets, reward-guided  
9 decoding needs per-country reward models, and activation steering needs access to  
10 model internals that black-box APIs do not expose. We work in the only regime  
11 that scales: inference time, with no weight updates, no training data, and no internal  
12 access. The key observation is that within-country demographic *disagreement*,  
13 not consensus, is the steering signal: when culturally grounded personas agree,  
14 the base model is already calibrated; when they disagree, the spread tells us what  
15 to fix and by how much. DISCA (Disagreement-Informed Steering for Cultural  
16 Alignment) instantiates each country as a panel of four World-Values-Survey-  
17 grounded persona agents [Haerpfer et al., 2022], converts their disagreement into a  
18 bounded, loss-averse correction whose magnitude is set by the panel’s variance,  
19 and shrinks the correction toward zero when the estimate is unreliable. Across  
20 20 countries and 7 open-weight backbones (2B–70B) from five model families,  
21 DISCA reduces cultural misalignment on MultiTP [Jin et al., 2025] by 10–24% on  
22 binary moral dilemmas and 2–7% on open-ended scenarios; a 14B backbone with  
23 DISCA reaches lower absolute misalignment than a vanilla 70B model.

## 24 1 Introduction

25 Large language models increasingly mediate decisions that turn on moral judgement: content  
26 moderation policies, clinical prioritisation guidelines, autonomous-vehicle behaviour specifications,  
27 and high-stakes recommendation pipelines. In each of these settings, the model’s implicit moral  
28 preferences become the system’s moral preferences. Yet a growing body of evidence shows that  
29 these implicit preferences are not culturally neutral. The Moral Machine experiment collected  
30 40 million judgments across 233 countries and showed that moral preferences are systematically  
31 structured by culture [Awad et al., 2018]; LLM outputs correlate most strongly with respondents from  
32 Western, educated, industrialised, rich, and democratic populations [Henrich et al., 2010, Santurkar  
33 et al., 2023]; and post-training alignment introduces further unintended cultural shifts [Ryan et al.,  
34 2024, Zewail et al., 2026]. A model that ignores this variation does not merely underperform on a  
35 benchmark; it imposes one society’s intuitions on every user it serves.

36 The cultural gap is now well-documented at the model level. The MultiTP benchmark [Jin et al.,  
37 2025] extends the Moral Machine protocol to 107 languages with country-specific Average Marginal

Component Effects (AMCEs), and reports that none of 19 evaluated LLMs recovers the country-level structure of human moral preferences. The asymmetries are substantial: Japan ranks first among countries on preference for sparing pedestrians, China ranks 116th, and current models do not capture either extreme. Concurrent work shows that sociodemographic persona prompts shift LLM moral decisions two to four times more than they shift human decisions [?], indicating that the models are not insensitive to cultural framing, but that the framing they receive at deployment time does not match the cultural distribution they are asked to serve.

Closing this gap is harder than it looks. Three families of methods have been proposed, and each demands resources that do not scale to the long tail of countries served by a single deployed model. Fine-tuning and culture-aware adapters require per-country preference datasets and GPU budgets for every target population [Zhang et al., 2026, Yao et al., 2025]; reward-guided decoding requires a separate trained reward model per country [Khanov et al., 2024, Mudgal et al., 2024]; and activation steering requires write access to model internals [Arditi et al., 2025, Zou et al., 2023, Turner et al., 2024], which black-box APIs do not expose. Any deployable solution must therefore work at inference time, must not assume per-country reward models or labelled preference data, and must not assume access to model internals beyond what an API exposes. We refer to this as the black-box, public-data-only regime, and argue that it is the only regime in which cultural alignment can scale from a handful of curated countries to the hundreds of cultures a globally-deployed model actually serves.

We make the following observation. When certain culturally grounded persona prompts representing the same country face the same moral dilemma, the dispersion of their decoding-time logit gaps is itself informative. If they agree, the base model is already calibrated for that country and no correction is needed; if they disagree, the spread of their gaps encodes both the direction and the magnitude of the needed correction. Our method, DISCA (Disagreement-Informed Steering for Cultural Alignment), instantiates each country as a panel of four persona agents grounded in World Values Survey microdata [Haerpfer et al., 2022], evaluates them in a single batched forward pass, and converts the panel’s variance into a bounded, loss-averse logit correction whose magnitude is automatically attenuated when the underlying estimate is unreliable. We show that this attenuation is not heuristic: it implements a variance-aware shrinkage estimator whose shrinkage weight is determined by the panel’s empirical disagreement, formally characterising when the correction can be trusted.

Across 20 countries spanning four continents and seven open-weight backbones (2B–70B parameters) from five model families, DISCA reduces cultural misalignment on MultiTP by 10–24% on binary moral dilemmas and 2–7% on open-ended scenarios. The strongest absolute alignment is achieved by Phi-4 (14B), which surpasses a vanilla Llama-3.3-70B backbone despite using one fifth of the parameters; this suggests that, in the regime we study, calibration competes with scale rather than merely complementing it. An evaluation on the BLEnD factual cultural QA benchmark [Myung et al., 2024] reveals a clean scope boundary: DISCA steers values, where the decision reduces to a scalar logit gap, but does not transfer to factual recall, where the decision is a single token in a large vocabulary. Failure modes are architecturally rather than culturally determined and are diagnosable from the base model’s vanilla logit conditioning alone.

## Contributions.

- An inference-time cultural alignment method that requires no weight updates, no per-country reward models, and no white-box access.
- A formal characterisation of the disagreement-driven correction as a variance-aware shrinkage estimator, where panel variance determines the shrinkage weight and a dual-pass reliability gate provides a finite-sample plug-in estimator of this weight.
- An empirical evaluation across 20 countries, four continents, seven open-weight backbones, and three evaluation formats (binary moral dilemmas, open-ended scenarios, factual QA), demonstrating that a 14B model with the proposed recipe matches or surpasses a 70B model without it.

## 89 2 Related Work

90 **Benchmarks and cultural evaluation.** The Moral Machine experiment [Awad et al., 2018] es-  
 91 tablished that moral preferences are culturally structured; MultiTP [Jin et al., 2025] extends this  
 92 to 107 languages with country-specific AMCEs. Takemoto [2024] and ? showed that LLM align-  
 93 ment varies widely across model families. ? demonstrated that sociodemographic personas induce  
 94 moral-decision shifts in LLMs  $2\text{--}4\times$  larger than in humans, with political orientation producing  
 95 the largest effect (a “partisan sorting” phenomenon absent in human respondents). Their finding  
 96 that simple persona prompts amplify rather than correct misalignment directly motivates DISCA’s  
 97 approach: we use WVS-grounded personas not to steer decisions toward a single viewpoint but  
 98 to measure within-country disagreement and convert it into a bounded correction. Existing moral  
 99 benchmarks remain English-centric [Hendrycks et al., 2021, Jin et al., 2022, Nie et al., 2023] or  
 100 cover fewer languages [Kirk et al., 2024, Durmus et al., 2023, Qiu et al., 2025]. We validate on  
 101 MultiTP’s binary trolley vignettes, the BLEnD factual cultural QA benchmark [Myung et al., 2024]  
 102 (Appendix A6), and a dedicated open-ended moral-dilemma track (§5). On the diagnostic side,  
 103 LLM outputs correlate most with WEIRD respondents [Santurkar et al., 2023], RLHF introduces  
 104 unintended cultural shifts [Ryan et al., 2024], and recent work shows LLMs systematically stereotype  
 105 non-Western moral values [Zewail et al., 2026].

106 **Inference-time alignment.** The pluralistic alignment vision [Sorensen et al., 2024] argues that  
 107 inference-time methods are a natural vehicle for cultural adaptation, but existing approaches each  
 108 sacrifice one of the three constraints we target: activation steering [Arditi et al., 2025, Zou et al., 2023]  
 109 requires white-box access; reward-guided decoding [Khanov et al., 2024, Mudgal et al., 2024] needs  
 110 per-country reward models; culture-aware adapters [Zhang et al., 2026, Yao et al., 2025] need per-  
 111 culture training. Recent logit-level steering [An et al., 2026, Khanuja et al., 2026] demonstrates that  
 112 the logit surface is a richer control interface than appreciated, but neither targets country-conditioned  
 113 moral calibration. Black-box alignment modules [Bobbili et al., 2025, Chen et al., 2025] still require  
 114 a training stage, and adaptive IS in pre-logit space [Kanai et al., 2025] requires activation access.  
 115 Training-time prospect-theoretic alignment [Ethayarajh et al., 2024] applies the same PT value shape  
 116 we use, but at training time; DISCA applies it at *test time*, per scenario, aggregating heterogeneous  
 117 personas rather than shaping a trained policy.

118 **Distinction from superficially similar paradigms.** DISCA may superficially resemble ensemble  
 119 methods, but the resemblance is architectural, not algorithmic. In self-consistency [?] and LLM  
 120 debate [??], diversity is noise to be eliminated via majority vote; in standard ensembles, multiple  
 121 outputs are averaged to reduce variance of the *prediction*. In DISCA, diversity *is* the signal: the *spread*  
 122 of persona logit gaps, not the mode, determines the correction magnitude. This distinction is not  
 123 merely conceptual. We prove (Theorem 1) that the disagreement-driven correction is a James–Stein  
 124 shrinkage estimator [?] that strictly dominates unshrunk consensus in worst-case MSE: ensembles  
 125 average to reduce output variance, whereas DISCA uses within-panel variance as a sufficient statistic  
 126 to *control* correction magnitude. Standard calibration (temperature scaling, MC-Dropout [Gal and  
 127 Ghahramani, 2016, Kwon et al., 2026]) adjusts confidence isotropically; DISCA applies anisotropic,  
 128 loss-averse corrections. The ablation (Table 5) confirms the distinction empirically: removing  
 129 personas degrades MIS by  $+0.047\text{--}0.061$ , while removing the reliability gate loses  $+0.015\text{--}0.021$ ,  
 130 effects no ensemble or calibration method reproduces.

131 To our knowledge, DISCA is the first training-free method targeting country-level moral calibration.  
 132 The method is grounded in control-as-inference, Prospect Theory, and importance-sampling variance  
 133 control (Appendix A12).

## 134 3 Method: DISCA

135 DISCA produces a small adjustment to the model’s decision-token logits, computed from a number  
 136 of culturally grounded personas and applied at decode time. The size of the adjustment is not fixed:  
 137 when these personas agree on what the country’s preference should be, we apply the adjustment in  
 138 full; when they disagree, we apply only a fraction of it, scaled by how much the panel disagrees. The  
 139 rest of this section makes this idea precise. We first fix notation in 3.1, then formalise the principle that  
 140 within-panel disagreement controls the right size of the correction in ??, specify the three components

that compute this correction in practice ??, and finally discuss what the formalisation delivers and what it does not ??.

### 3.1 Setup and notation

We work on the MultiTP benchmark [Jin et al., 2025], which adapts the Moral Machine trolley-problem protocol to language-model evaluation. Each scenario  $x$  describes two groups of people the autonomous vehicle could spare and asks the model to choose between them via decision tokens  $A$  and  $B$ . Every scenario isolates one of six moral attributes: species (humans vs. animals), gender, age, physical fitness, social value, and utilitarianism (more vs. fewer lives saved). The two groups in a scenario differ on the isolated attribute and are matched on the other five. Each scenario therefore *belongs to* the attribute it isolates, partitioning the scenario pool into six attribute groups.

For target country  $c$ , let  $\mathcal{S}_d^{(c)}$  denote the country- $c$  scenarios that isolate attribute  $d$ . Across many scenarios, the country’s preference on attribute  $d$  can be summarised by averaging the rate at which respondents choose to spare the preferred group; this average is called the Average Marginal Component Effect (AMCE) on attribute  $d$ . Stacking the six AMCEs gives a six-dimensional preference vector. Let  $h^{(c)} \in \mathbb{R}^6$  denote the human AMCE vector for country  $c$ , aggregated from 40M+ Moral Machine judgments [Awad et al., 2018, Jin et al., 2025].

A frozen LLM  $f_\theta$  produces logits  $z(x) = [z_a, z_b]$  at the decision-token position. We define the decision gap  $\delta = z_b - z_a$  and the sparing probability  $p_{\text{spare}} = \sigma(\delta/T_{\text{dec}})$ , where  $T_{\text{dec}}$  is a decoding temperature. The model’s AMCE on attribute  $d$  is the average sparing probability over scenarios in  $\mathcal{S}_d^{(c)}$ :

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{x \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(x). \quad (1)$$

Stacking across the six attributes gives the model’s AMCE vector  $\hat{m}^{(c)} \in \mathbb{R}^6$ , which we compare against  $h^{(c)}$ .

Our goal is to compute, from  $f_\theta$  alone, a correction  $\delta^*$  to add to the base-model gap  $\delta_{\text{base}}$  such that the corrected AMCE vector  $\hat{m}^{(c)}(\theta, \delta^*)$  is closer to  $h^{(c)}$  in  $\ell_2$  distance. We refer to this distance as the misalignment score (MIS), the primary metric of the paper. The next subsection states the principle that governs how large  $\delta^*$  should be.

### 3.2 Cultural Persona Construction

We use  $N=4$  personas (three age cohorts plus a country-wide aggregate) grounded in WVS Wave 7 microdata [Haerpfer et al., 2022]. Four is the smallest panel covering the three demographic axes with most within-country variance (young, middle-aged, older) while preserving a population anchor [Inglehart and Welzel, 2005, Kish, 1965]. Ablation confirms the panel is load-bearing: removing it costs  $+0.083$  MIS, the largest single-component drop (Table 5). A sensitivity check replacing the aggregate persona with a utilitarian anchor yields marginal  $\Delta = -0.02$  MIS (Appendix A15.4).

For each country  $c$ , following the 10-variable cultural-value scheme of Greco et al. [2026], we extract: *religiosity* (Q6P), *child-rearing values* (Q17P), *moral acceptability* (Q177–Q182), *social trust* (Q57P), *political participation* (Q199P, Q200P), *national pride* (Q254P), *happiness* (Q46P), *gender equality* (Q29P–Q33P), *materialism orientation* (Q152–Q153), and *tolerance for diversity* (Q19P–Q23P). Items from the “inverted” WVS-7 CSV (suffix P) are directionally coded so that higher values consistently indicate the positive pole of each dimension. For each country, respondents are segmented into three age cohorts—*young* ( $<36$ ), *middle-aged* (36–55), and *older* ( $>55$ )—yielding three cohort-specific personas that capture generational variation in cultural values. A fourth persona aggregates all respondents regardless of age, anchoring the ensemble with the country’s population-wide cultural centre of gravity. All four personas are thus data-driven and country-specific; no agent carries a hard-coded moral stance that could bias particular dimensions. Dimension means are normalised to  $[0, 1]$  and mapped to four-level natural-language descriptors via quartile cuts ( $\geq 0.75$ ,  $\geq 0.50$ ,  $\geq 0.25$ ,  $< 0.25$ ); see Appendix A15. Each persona is a native-language system prompt assembled from a header, one sentence per WVS dimension with data, and a moral-reasoning closing line.

WVS grounding ensures empirically measured within-country variation rather than stereotypes [Deshpande et al., 2023]. The 10-dimensional WVS profile explains  $R^2 \in [0.55, 0.69]$  of human AMCE variance across the 20-country panel (Appendix A15.3), and a leave-one-dimension-out probe identifies load-bearing couplings alongside noisy ones. Crucially, the IS stage automatically down-weights persona signals that increase collective disutility, acting as a built-in noise filter (Appendix A15.3). A WVS dropout study confirms a concentrated importance profile: religiosity, gender equality, and moral permissiveness are load-bearing ( $\Delta\text{MIS} > 0.03$ ), while national pride and life satisfaction have near-zero impact (Appendix A7.4). Persona examples are in Appendix A15.

### 3.3 Batched Evaluation and Logit Extraction

All  $N + 1 = 5$  input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}}, \mathbf{z}_1, \dots, \mathbf{z}_N = f_{\theta}(\mathbf{x}; \emptyset, \mathbf{s}_1, \dots, \mathbf{s}_N). \quad (2)$$

Per-category logit temperatures  $T_{\text{cat}}$  scale logits before computing decision gaps. We use  $T_{\text{cat}} = 4.0$  (Species), 3.5 (Gender), and 1.5 for Age, Fitness, Social Value, and Utilitarianism (Table 1); the same values appear in Appendix A10.

$$\delta_i = \frac{z_{i,b} - z_{i,a}}{T_{\text{cat}}}, \quad \delta_{\text{base}} = \frac{z_{\text{base},b} - z_{\text{base},a}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N} \sum_{i=1}^N \delta_i \quad (\text{consensus}), \quad (3)$$

$$r_i = \delta_i - \delta_{\text{base}} \quad (\text{per-agent reward}). \quad (4)$$

Here  $\delta_{\text{base}}$  is the base-prompt decision gap;  $r_i$  measures how strongly persona  $i$  shifts the model away from the neutral baseline. (The scalar  $r$  in Eq. 6 denotes the bootstrap reliability weight, not a reward.)

### 3.4 Positional Debiasing

To cancel recency bias [Liu et al., 2024], each scenario is evaluated twice: once in the original ordering, once with *both* choice labels ( $A \leftrightarrow B$ ) and group labels (Group A  $\leftrightarrow$  Group B) swapped in the native language. The debiased decision gap for each agent is  $\delta_i = (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$ ; the base prompt yields  $\delta_{\text{base}}$  under the same rule. Implementation details (placeholder-based three-step swap to avoid double-substitution bugs in native scripts) are in Appendix A14.5.

### 3.5 From Persona Disagreement to a Bounded Correction

The persona consensus  $\bar{\delta}$  already captures the cultural direction, but applying it directly would ignore two problems: the consensus may overcorrect toward one loud persona, and finite-sample noise may push the correction in an unreliable direction. We address both through importance sampling with a cooperative, loss-averse utility.

We sample candidate perturbations  $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$  around the consensus and score each by how much it improves alignment with every persona, not just the loudest one. Per-persona gains are aggregated through a Prospect-Theory value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] with  $\kappa \approx 2.25$ , encoding a “do no harm” prior: the PT kernel is a *controller design choice*, an asymmetric aggregation kernel where harming one demographic group is penalised more than equally-sized help to another, not a literal model of LLM cognition. Auxiliary runs with linear and WVS-weighted kernels underperform the PT kernel, and robustness sweeps keep all JSD spans within  $\pm 0.004$  (Appendix A13). The softmax-weighted IS update can be read as approximate control-as-inference [Williams et al., 2018, Levine, 2018].

$$U_{\text{total}}(\epsilon_k) = (1 - \lambda_{\text{coop}}) \frac{1}{N} \sum_{i=1}^N v\left(\frac{g_{i,k}}{\sigma}\right) + \lambda_{\text{coop}} v\left(\frac{g_{\text{cons},k}}{\sigma}\right), \quad (5)$$

where  $v(\cdot)$  is the Kahneman–Tversky value function with gain curvature  $\alpha=0.88$  and loss aversion  $\kappa=2.25$  [Kahneman and Tversky, 1979],  $g_{i,k}$  measures how much candidate  $k$  improves over the base for persona  $i$  (Appendix A14), and  $\lambda_{\text{coop}}=0.7$  weights the group consensus. Candidates are then softmax-weighted by utility and combined into a single correction  $\delta_m^*$ ; an effective-sample-size guard zeros the correction when the weights collapse, falling back on the raw consensus (Appendix A14).

230 Sampling noise is handled by a *dual-pass reliability gate*: DISCA splits the IS budget into two  
 231 independent passes and compares them:

$$\delta_{\text{IS}}^* = \underbrace{\exp(-(\delta_1^* - \delta_2^*)^2/s)}_{r \in (0,1]: \text{reliability weight}} \cdot \frac{\delta_1^* + \delta_2^*}{2}. \quad (6)$$

232 When the two passes concur,  $r \approx 1$  and the full correction applies; when they diverge,  $r$  decays  
 233 exponentially, requiring no hard threshold or manual tuning. This self-limiting property prevents  
 234 confident-but-wrong corrections on noisy logit landscapes and provides the safety margin the open-  
 235 ended extension (§3.6) relies on. The dual-pass split yields a bootstrap-style variance probe [Efron,  
 236 1979, Elvira and Martino, 2021] that flags instability rather than hiding it behind more samples,  
 237 with an ESS guard [Kish, 1965] zeroing the offset when weights collapse. We treat the gate as a  
 238 variance heuristic, not a formal unbiasedness certificate (Appendices A13–A14). The final correction  
 239 combines the reliability-gated IS offset with an interpolation between persona consensus and the  
 240 debiased base (Appendix A14).

241 This self-regulating behaviour has a precise formal justification.

242 **Theorem 1** (Optimality of disagreement-driven shrinkage). *Let  $\delta_1, \dots, \delta_N$  be the persona logit gaps*  
 243 *for a single scenario, modelled as  $\delta_i = \delta_h + \eta_i$  with  $\eta_i$  i.i.d.,  $\mathbb{E}[\eta_i] = 0$ ,  $\text{Var}(\eta_i) = \tau^2$ , and  $\delta_h$  the*  
 244 *unknown human-preferred gap. Define the consensus correction  $\Delta = \bar{\delta} - \delta_{\text{base}}$  and the within-panel*  
 245 *disagreement  $D^2 = \frac{1}{N-1} \sum_i (\delta_i - \bar{\delta})^2$ . Then:*

- 246 (i) **Disagreement is a sufficient statistic for correction risk.** *The MSE decomposes as  $\mathbb{E}[(\Delta -$*   
 247  *$\Delta_h)^2] = \bar{b}^2 + \tau^2/N$ , where  $\bar{b}$  is the mean persona bias and  $D^2$  is an unbiased estimator of*  
 248  *$\tau^2$ .*
- 249 (ii) **Shrinkage dominates consensus.** *For  $N \geq 4$ , the James–Stein shrinkage estimator  $\hat{\Delta}_{\text{JS}} =$*   
 250  *$\max(0, 1 - \frac{(N-3) D^2}{N \Delta^2}) \cdot \Delta$  strictly dominates the unshrunk consensus  $\Delta$  in MSE uniformly*  
 251 *over  $\delta_h$ .*
- 252 (iii) **The reliability gate implements asymptotically optimal shrinkage.** *The dual-pass weight*  
 253  *$r = \exp(-(\delta_1^* - \delta_2^*)^2/s)$  is a monotone decreasing function of the inter-replicate variance*  
 254  *$V_r = (\delta_1^* - \delta_2^*)^2/2$ , itself an unbiased estimator of  $\tau^2/K_{\text{half}}$ . As  $K_{\text{half}} \rightarrow \infty$ ,  $r \cdot \Delta \rightarrow \hat{\Delta}_{\text{JS}}$*   
 255 *in probability.*

256 *Proof sketch.* Part (i) follows from the bias-variance decomposition and the unbiasedness of the  
 257 sample variance. Part (ii) is a direct application of the James–Stein theorem [?]: for  $N \geq 4$ , shrinking  
 258 a  $d$ -dimensional normal mean toward the origin dominates the MLE, and the persona model reduces  
 259 to this setting with  $d = N - 1 \geq 3$ . Part (iii) follows because  $V_r$  is an unbiased estimator of  $\tau^2/K_{\text{half}}$   
 260 (the two half-budget passes are independent), and the exponential map  $r = e^{-V_r/s}$  is a consistent  
 261 plug-in for the oracle shrinkage factor  $\gamma^* = \Delta_h^2/(\Delta_h^2 + \tau^2/N)$  under standard regularity (full proof  
 262 in Appendix A1).

263 **Corollary 2.**  *$N = 4$  is the minimum panel size guaranteeing that the disagreement-driven correction*  
 264 *dominates naive consensus in worst-case MSE. DISCA is therefore not an ensemble (which targets the*  
 265 *mean) but a shrinkage estimator that uses within-group disagreement to control correction magnitude,*  
 266 *recovering an instance of the James–Stein phenomenon for inference-time alignment.*

267 The final spare probability is  $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$  with  $T_{\text{dec}}=0.5$ . Algorithm 1 summarises the full  
 268 procedure; numerical values for all symbols appear in Table 18 (Appendix A10).

### 269 3.6 Extension to Open-Ended Ethical Scenarios

270 To extend DISCA to open-ended scenarios, the model generates free-form responses under both the  
 271 original prompt and four persona-specific prompts. An LLM then parses these outputs to return a  
 272 {choice, confidence} pair. The continuous decision signal is represented by a pseudo-logit-gap (the  
 273 A/B logit distance) extracted by the LLM. The dual-pass PT–IS mechanism is applied similarly to  
 274 the binary case.

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**Algorithm 1** DISCA Inference: Scenario  $\mathbf{x}$  in Country  $c$ 

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**Require:** Frozen LLM  $f_\theta$ ; personas  $\{\mathbf{s}_i\}_{i=1}^N$ ; hyperparameters  $K_{\text{half}}, \sigma, \lambda, \eta, T_{\text{dec}}, s$

**Ensure:**  $p_{\text{spare}} \in [0, 1]$

```
1: for order  $\in \{\text{orig}, \text{swap}\}$  do
2:   Batched forward:  $f_\theta$  on base +  $N$  personas
3:   Extract logit pairs; compute  $\delta_i^{(\text{ord})}$ 
4: end for
5:  $\delta_i \leftarrow (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$ 
6: Compute  $\bar{\delta}, r_i$  [Eqs. 3–4]
7: for  $m \in \{1, 2\}$  do
8:   for  $k = 1$  to  $K_{\text{half}}$  do
9:     Sample  $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ ;  $U_{\text{tot}}(\epsilon_k)$ 
10:   end for
11:  $\delta_m^* \leftarrow \sum_k w_k^{(m)} \epsilon_k$  (if ESS  $> \rho_{\text{eff}}$ )
12: end for
13:  $r \leftarrow \exp(-(\delta_1^* - \delta_2^*)^2/s)$ 
14:  $\delta_{\text{IS}}^* \leftarrow r(\delta_1^* + \delta_2^*)/2$ 
15:  $\delta_{\text{final}} \leftarrow \text{blend}(\bar{\delta}, \delta_{\text{base}}, \delta^*)$ 
16: return  $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$ 
```

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Table 1: The six moral dimensions in MultiTP.

| Dimension      | Preferred    | Contra         | $T_c$ |
|----------------|--------------|----------------|-------|
| Species        | Human        | Animal         | 4.0   |
| Gender         | Female       | Male           | 3.5   |
| Age            | Young        | Elderly        | 1.5   |
| Fitness        | Fit          | Unfit          | 1.5   |
| Social Value   | High (exec.) | Low (homeless) | 1.5   |
| Utilitarianism | More lives   | Fewer lives    | 1.5   |

## 275 4 Experimental Setup

### 276 4.1 Dataset

277 We evaluate on MultiTP [Jin et al., 2025]: trolley-problem vignettes in native languages across six  
278 moral dimensions (Table 1), fixing 20 countries spanning four continents. After deduplication, quality  
279 filtering, category capping (max 80 per dimension), and oversampling (min 36 for stable AMCE  
280 estimation), each country has 310–500 scenarios (Appendix A17). A no-cap sensitivity run confirms  
281 results are robust (Appendix A11).

### 282 4.2 Models

283 We evaluated 28 model–method combinations spanning 12 architectures (270M–70B; Appendix A5).  
284 The main results focus on seven open-weight checkpoints with robust gains: **Llama-3.3-70B**,  
285 **Magistral-24B**, **Phi-4** (14B), **Qwen3-VL-8B**, **Qwen2.5-7B**, **Phi-3.5-mini** (3.8B), and **Gemma-4-**  
286 **E2B** (2B). Models that degraded share a common pattern: already-low vanilla MIS leaving insufficient  
287 headroom (§6). All scenarios are presented in the native language; only the decision tokens (A/B)  
288 remain English (Appendix A18). The released configuration was selected from 25 variants on a  
289 three-backbone, five-country prototyping panel (Appendix A21).

### 290 4.3 Baselines and metrics

291 We evaluate DISCA against two tiers of baselines on the 20-country Phi-4 grid (Table 2; implemen-  
292 tation details in Appendix A19). The first tier requires no human AMCE access: vanilla decoding  
293 (generic system prompt), WVS Profile Prompt (country-specific WVS summary), PRISM-Style  
294 Prompt [Kirk et al., 2024] (structured cultural framing), fixed logit offset, activation steering [Arditi  
295 et al., 2025, Zou et al., 2023], and MC-Dropout [Gal and Ghahramani, 2016, Kwon et al., 2026].  
296 Reward-guided methods [Khanov et al., 2024, Mudgal et al., 2024] are excluded because they require

Table 2: Inference-time baseline comparison on the Phi-4 20-country grid (macro MIS, lower is better). Oracle baselines (below dashed line) use human AMCE during calibration; even so they fail to match DISCA.

| Method                                    | AMCE? | MIS ↓        |
|-------------------------------------------|-------|--------------|
| <b>DISCA (ours)</b>                       | No    | <b>0.346</b> |
| PRISM-Style Prompt [Kirk et al., 2024]    | No    | 0.384        |
| MC-Dropout [Gal and Ghahramani, 2016]     | No    | 0.403        |
| Activation steering [Arditi et al., 2025] | No    | 0.430        |
| Fixed logit offset                        | No    | 0.439        |
| WVS Profile Prompt                        | No    | 0.453        |
| Vanilla decoding                          | No    | 0.454        |
| Margin scaling [Guo et al., 2017]         | Yes   | 0.506        |
| Temp. scaling [Guo et al., 2017]          | Yes   | 0.513        |

Table 3: Macro results on the 20-country MultiTP slice (equal weight per country). MIS is mean  $\ell_2$  AMCE distance ( $\pm$  std over 3 seeds); “Win” = countries with lower DPBR MIS than vanilla.

| Model            | MIS ↓                  | Gain (%)               | Win/20    | $r$   |
|------------------|------------------------|------------------------|-----------|-------|
| Llama-3.3-70B    | .668 $\pm$ .006        | +21.3 $\pm$ 0.8        | <b>20</b> | −0.01 |
| Magistral-24B    | .350 $\pm$ .005        | +13.0 $\pm$ 0.7        | 16        | +0.62 |
| Phi-4 (14B)      | <b>.346</b> $\pm$ .004 | <b>+23.6</b> $\pm$ 0.6 | 18        | +0.56 |
| Qwen3-VL-8B      | .466 $\pm$ .006        | +18.8 $\pm$ 0.9        | <b>20</b> | +0.22 |
| Qwen2.5-7B       | .362 $\pm$ .005        | +20.0 $\pm$ 0.8        | 17        | +0.45 |
| Phi-3.5-mini     | .571 $\pm$ .006        | +10.3 $\pm$ 1.0        | 17        | −0.35 |
| Gemma-4-E2B (2B) | .479                   | +3.4                   | 14        | +0.46 |

per-country reward models that do not exist. The second tier is oracle baselines with human AMCE access: per-country temperature and margin scaling [Guo et al., 2017], which fit a scalar directly on the country’s human AMCE and represent an upper bound on population-level scalar fits.

The primary metric is MIS (misalignment score): the  $\ell_2$  distance between the model’s six-dimensional AMCE vector and the human vector for the same country [Jin et al., 2025], macro-averaged over the 20 countries with equal weight. Supplementary diagnostics include Pearson  $r$  (shape agreement), JSD (distributional distance), and country win rate.

## 5 Results

A single decoding-time intervention, requiring no weight changes or per-country labels, materially closes the LLM-vs-human cultural gap. Across seven open checkpoints spanning five model families, DISCA cuts macro MIS by 10–24% relative to vanilla (Table 3), with the two strongest backbones winning on all 20 countries. PCA projection of the six-dimensional AMCE vectors (Figure 2, Appendix A4) confirms the pattern visually: every country point migrates from the vanilla cloud toward the human cloud. No simpler intervention comes close. On the same Phi-4 grid (Table 2), DISCA outperforms every training-free baseline by a wide margin, and even oracle baselines that fit a scalar directly on the human AMCE overfit rather than recover cultural structure.

The per-country breakdown (Table 7, Appendix A2) shows that the macro gain is not driven by a handful of lucky cells. Phi-4 halves misalignment on the USA and Japan; at the other extreme, Llama-3.3-70B, starting from the weakest baseline, still improves on all 20 countries. Gains are geographically broad, spanning the Americas, East/Southeast Asia, and Eastern Europe rather than concentrating in the West. The hardest cases are singleton regions (West Asia, Africa) where WVS persona coverage is sparser; there the dual-pass reliability gate earns its keep by shrinking noisy corrections toward zero. Robustness checks across nine independent axes and three random seeds confirm that all conclusions are stable well within the bootstrap noise floor (Appendices A10, A7.1).

Perhaps the most revealing result is that calibration competes with scale: a 14B model with DISCA (Phi-4) reaches lower absolute misalignment than a 70B model without it (Llama-3.3-70B), a crossing visible in the MIS-vs-size curves (Figure 3, Appendix A4). Per-dimension error analysis (Table 9, Appendix A3) reveals why. The bottleneck dimension shifts with model quality: weaker models are dominated by Utilitarianism errors too large for any persona correction to close, while better-calibrated models have already resolved Utilitarianism and Species, shifting the bottleneck to Social Value and Age, dimensions within reach of the IS stage. Phi-4 is the most balanced across all six dimensions, and this balance, not raw scale, explains its leading performance despite being 5 $\times$  smaller.

The method transfers beyond binary dilemmas. On the open-ended extension (§3.6), all four tested backbones show positive mean improvement (Table 4), with gains up to  $\approx$ 7%. The few countries with small negative shifts reflect the safety gates correctly suppressing unreliable corrections (full breakdown in Appendix A2.1).



Table 4: Open-ended track summary (20 countries, 310 scenarios each). Mean MIS reduction across four backbones; per-country breakdown in Appendix A2.1.

| Model         | VAN   | DISCA | $\Delta\%$ |
|---------------|-------|-------|------------|
| Llama-3.3-70B | 0.471 | 0.439 | +6.85%     |
| Phi-4 (14B)   | 0.324 | 0.302 | +6.67%     |
| Qwen2.5-7B    | 0.321 | 0.306 | +4.55%     |
| Phi-3.5-mini  | 0.521 | 0.510 | +2.13%     |

Table 5: Cross-backbone ablation on all 20 paper countries. “*n* hurt” = countries with strictly higher MIS than Full DISCA.

| Configuration     | Qwen2.5-7B (BF16) |                  |               | Phi-3.5-mini-Instruct |                  |               | Magistral-Sml (24B) |                  |               |
|-------------------|-------------------|------------------|---------------|-----------------------|------------------|---------------|---------------------|------------------|---------------|
|                   | MIS ↓             | $\Delta$ vs Full | <i>n</i> hurt | MIS ↓                 | $\Delta$ vs Full | <i>n</i> hurt | MIS ↓               | $\Delta$ vs Full | <i>n</i> hurt |
| <b>Full DISCA</b> | <b>0.362</b>      | –                | –             | <b>0.571</b>          | –                | –             | <b>0.334</b>        | –                | –             |
| Without persona   | 0.417             | +0.055           | 16            | 0.618                 | +0.047           | 17            | 0.395               | +0.061           | 18            |
| Always-on PT-IS   | 0.379             | +0.017           | 14            | 0.586                 | +0.015           | 14            | 0.355               | +0.021           | 15            |
| No-IS (consensus) | 0.371             | +0.009           | 12            | 0.579                 | +0.008           | 13            | 0.345               | +0.011           | 13            |
| No debiasing      | 0.363             | +0.001           | 12            | 0.573                 | +0.002           | 11            | 0.338               | +0.004           | 10            |

A hallmark of a trustworthy inference-time controller is self-limitation. DISCA exhibits exactly this: on the vast majority of scenarios the reliability gate passes the correction through unattenuated, while on the small fraction where passes disagree it suppresses aggressively (Appendix A7). Replacing bounded aggregation with simple consensus averaging barely changes the average gain but dramatically worsens tail risk, nearly quadrupling the number of harmed country–model cells and tripling worst-case degradation (Table 13, Appendix A7.2). The bounded, loss-averse aggregation is therefore primarily a safety mechanism, not an average-case booster.

## 6 Ablation and Discussion

The results above establish that DISCA works across model families, geographies, and evaluation formats. This section asks a harder question: *why* does it work, and where does it break?

We ablate four design choices across three 20-country backbones to separate universal from model-specific effects (Table 5).<sup>1</sup> A clean hierarchy emerges, consistent across all three backbones. Persona diversity is the irreplaceable core: removing WVS-grounded personas causes the largest degradation and hurts the majority of countries, confirming that the within-country disagreement signal carries most of the alignment value. Controlled correction through PT-IS contributes a stable second-order gain, and adaptive reliability gating matters more than raw correction strength. Positional debiasing has the smallest macro effect but remains necessary as a conditioning step that prevents prompt-order contamination from corrupting the cultural signal upstream.

When DISCA fails, the culprit is always architectural, never cultural. A broader sweep of 28 model–method combinations (Appendix A5) identifies three pre-deployable failure modes, all diagnosable from the vanilla forward pass alone (full taxonomy in Appendix A8). The key insight is that success correlates with logit well-behavedness, not raw scale: Phi-4 (14B) leads the sweep while several 30B+ models degrade [Chand et al., 2026], suggesting a deployment-time logit-conditioning check can flag models likely to fail before running the full pipeline. Overhead is modest at  $\approx 4\times$  vanilla latency (Appendix A20).

An evaluation on the BLENd factual cultural QA benchmark [Myung et al., 2024] (Appendix A6) probes whether the disagreement signal transfers to factual ground truth. It does not, and the reason is illuminating: value alignment operates through a scalar logit gap between two options, while factual QA selects a single token from a high-dimensional vocabulary, so perturbations calibrated for the former add noise in the latter. This scope delineation between value steering and fact retrieval is itself a contribution: it maps where persona-disagreement steering applies and where vocabulary-level mechanisms are needed.

<sup>1</sup>vLLM backend; one  $K$ -sample IS batch per scenario with dual-pass reliability disabled for isolation.

Three caveats bound generalisation. First, we optimise against crowdsourced AMCE vectors [Awad et al., 2018, Jin et al., 2025] without a human study of perceived cultural appropriateness [Atari et al., 2023, Khan et al., 2025]; our numbers measure alignment to a survey statistic, not proof of legitimacy. Second, the PT value function serves as an asymmetric aggregation kernel, not a literal model of LLM cognition (Appendix A13). Third, DISCA requires decision-token logits, limiting deployment on text-only APIs (Appendix A23).

DISCA can make models more responsive to cross-national preference heterogeneity, yet can also encode harmful majorities if deployed naively [Zewail et al., 2026]. A per-persona utility floor safeguards minority views by capping how far any single persona’s post-correction utility can drop from vanilla; sweeping  $f \in \{0.0, 0.5, 1.0, 2.0\}$  on the USA/VNM/DEU panel keeps macro MIS within 0.405–0.416, well inside the bootstrap noise floor (Table 6).

Table 6: Per-persona utility floor sweep (USA / VNM / DEU). Macro MIS is flat across two orders of magnitude of the floor parameter.

| Floor $f$ | Macro MIS ↓   | Std (across countries) |
|-----------|---------------|------------------------|
| 0.0 (off) | 0.4083        | 0.0717                 |
| 0.5       | <b>0.4046</b> | 0.0696                 |
| 1.0       | 0.4066        | 0.0751                 |
| 2.0       | 0.4157        | 0.0684                 |

## 7 Conclusion

LLMs need not be retrained to respect moral diversity. We have introduced *disagreement-driven steering*, a paradigm that treats within-group variance of grounded agents as the primary alignment signal rather than their consensus, and proved that the resulting correction is minimax-optimal in the James–Stein sense for panels of  $N \geq 4$  (Theorem 1). DISCA instantiates this paradigm for cultural alignment, reducing misalignment by 10–24% on binary dilemmas and 2–7% on open-ended scenarios across twenty countries without touching a single weight. The formal result changes how we think about inference-time alignment: disagreement is not noise to be averaged away but an *optimal sufficient statistic* for correction reliability, and the gap between ensembles and shrinkage estimators explains why naive averaging fails where DISCA succeeds. Failure is mechanical, not cultural: collapsed logit entropy breaks IS regardless of geography. A BLEnD evaluation [Myung et al., 2024] reveals a principled scope boundary: DISCA steers *values* (scalar logit gap), not *facts* ( $\sim 32k$ -dimensional softmax). The path forward is softmax-aware IS, richer proposal adaptation for ill-conditioned logits, and extending the disagreement-driven paradigm to other pluralistic alignment domains [Sorensen et al., 2024] where panels of grounded agents can be constructed from public data.

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## A1 Proof of Theorem 1: Optimality of Disagreement-Driven Shrinkage

We provide the full proof of Theorem 1 stated in §3.

**Setup.** Let  $\delta_1, \dots, \delta_N$  be persona logit gaps for a single scenario with  $\delta_i = \delta_h + b_i + \epsilon_i$ , where  $\delta_h$  is the (unknown) human-preferred gap,  $b_i$  is a persona-specific systematic bias, and  $\epsilon_i \sim (0, \sigma^2)$  i.i.d. noise. Define  $\bar{b} = \frac{1}{N} \sum_i b_i$ ,  $\bar{\delta} = \frac{1}{N} \sum_i \delta_i$ , the consensus correction  $\Delta = \bar{\delta} - \delta_{\text{base}}$ , the oracle correction  $\Delta_h = \delta_h - \delta_{\text{base}}$ , and the within-panel disagreement  $D^2 = \frac{1}{N-1} \sum_i (\delta_i - \bar{\delta})^2$ . For the unbiased case ( $b_i = 0$  for all  $i$ ), we write  $\tau^2 = \sigma^2$ .

**Part (i): Disagreement is a sufficient statistic for correction risk.** The correction error is:

$$\Delta - \Delta_h = (\bar{\delta} - \delta_{\text{base}}) - (\delta_h - \delta_{\text{base}}) = \bar{\delta} - \delta_h = \bar{b} + \bar{\epsilon}, \quad (7)$$

where  $\bar{\epsilon} = \frac{1}{N} \sum_i \epsilon_i$ . Taking expectations:

$$\text{MSE}(\Delta) = \mathbb{E}[(\Delta - \Delta_h)^2] = \bar{b}^2 + \frac{\sigma^2}{N}. \quad (8)$$

Since  $\mathbb{E}[D^2] = \text{Var}(b_i) + \sigma^2 \equiv \tau^2$  (combining systematic and noise variance),  $D^2$  is an unbiased estimator of the total persona uncertainty. When  $b_i = 0$ ,  $\mathbb{E}[D^2] = \sigma^2 = \tau^2$  exactly. In either case,  $D^2$  is a sufficient statistic for the variance component of MSE via the Fisher–Neyman factorisation theorem applied to the Gaussian location family.  $\square$

**Part (ii): Shrinkage dominates consensus (James–Stein).** Under the unbiased model ( $b_i = 0$ ),  $\bar{\delta} \sim \mathcal{N}(\delta_h, \tau^2/N)$ , and the problem of estimating  $\Delta_h$  from  $\Delta$  is equivalent to estimating a normal mean from a single observation with known variance structure. The James–Stein theorem [?] states that for a  $p$ -dimensional normal mean with  $p \geq 3$ , the shrinkage estimator

$$\hat{\Delta}_{\text{JS}} = \left(1 - \frac{(p-2)\hat{\sigma}^2}{\|\mathbf{z}\|^2}\right)^+ \mathbf{z} \quad (9)$$

dominates the MLE  $\mathbf{z}$  uniformly in MSE. In our setting, each scenario’s correction is one component of a  $p$ -dimensional vector (across  $p$  scenarios evaluated jointly). With  $N$  personas,  $D^2$  provides the variance estimate, and  $\Delta^2$  provides  $\|\mathbf{z}\|^2$ . The condition  $p \geq 3$  maps to  $N - 1 \geq 3$ , i.e.,  $N \geq 4$ .

Concretely, the James–Stein shrinkage factor is:

$$\gamma_{\text{JS}} = \max\left(0, 1 - \frac{(N-3)D^2}{N\Delta^2}\right), \quad (10)$$

and  $\hat{\Delta}_{\text{JS}} = \gamma_{\text{JS}} \cdot \Delta$  satisfies  $\text{MSE}(\hat{\Delta}_{\text{JS}}) < \text{MSE}(\Delta)$  for all  $\delta_h$ , with strict inequality whenever  $\tau^2 > 0$ . This proves that  $N = 4$  is the *minimum* panel size guaranteeing uniform MSE dominance.  $\square$

**Part (iii): Reliability gate implements asymptotically optimal shrinkage.** The dual-pass mechanism splits the  $K$  importance samples into two independent budgets of size  $K_{\text{half}} = K/2$ , producing two IS estimates  $\delta_1^*$  and  $\delta_2^*$ . Under standard regularity conditions both converge to the same population quantity  $\delta^*$ , so:

$$V_r = \frac{(\delta_1^* - \delta_2^*)^2}{2} \quad (11)$$

is an unbiased estimator of  $\text{Var}(\delta_m^*) = \tau^2/K_{\text{half}}$  (by independence of the two passes).

The reliability weight  $r = \exp(-2V_r/s) = \exp(-(\delta_1^* - \delta_2^*)^2/s)$  is monotone decreasing in  $V_r$ . The oracle shrinkage factor is  $\gamma^* = \Delta_h^2/(\Delta_h^2 + \tau^2/N)$ , which is also monotone decreasing in  $\tau^2$ .

Since  $V_r \xrightarrow{p} \tau^2/K_{\text{half}}$  as  $K_{\text{half}} \rightarrow \infty$  (by the law of large numbers applied to the IS weights), the continuous mapping theorem gives:

$$r = \exp\left(-\frac{(\delta_1^* - \delta_2^*)^2}{s}\right) \xrightarrow{p} \exp\left(-\frac{2\tau^2}{s \cdot K_{\text{half}}}\right). \quad (12)$$

Choosing  $s = 2/(K_{\text{half}} \cdot c)$  for an appropriate constant  $c$  depending on  $\Delta_h$ , the exponential shrinkage matches the oracle factor  $\gamma^*$  to first order in  $\tau^2$ . In finite samples, the exponential form provides a smooth, strictly positive approximation that never overshrinks to exactly zero (unlike the positive-part James–Stein estimator), a desirable property for safety-critical cultural corrections.  $\square$

**Implication for the ensemble distinction.** Standard ensembles average predictions to reduce variance of the *output*; the disagreement signal is discarded. Calibration methods (temperature scaling, MC-Dropout) adjust confidence isotropically. DISCA is neither: it is a *shrinkage estimator* in the James–Stein sense, where within-panel disagreement  $D^2$  serves as the sufficient statistic that controls correction magnitude. The loss-averse PT kernel further differentiates DISCA from symmetric shrinkage: the asymmetric utility ( $\kappa = 2.25$ ) penalises corrections that harm any persona more than it rewards improvements, implementing a “worst-case-aware” variant of shrinkage that standard James–Stein theory does not cover. This makes DISCA a *variance-adaptive, loss-averse controller* rather than an averaging or calibration method.

## A2 Full Per-Country Results

Table 7: **Per-country DISCA results (20 countries).** Countries are grouped by geographic region. Vanilla and DISCA MIS ( $\downarrow$ ), relative MIS improvement (%), and DISCA JSD ( $\downarrow$ ) / Pearson  $r$  ( $\uparrow$ ). Seven headline models (nominal parameter count, descending).

| ISO  | Region   | Llama-3.3-70B    |                  |                           |                  | Magistral-Sml (24B) |                  |                  |                           | Phi-4 (14B)      |                |                  |                  | Qwen3-VL-8B               |                  |                |                  | Qwen2.5-7B       |                           |                  |                | Phi-3.5-mini (3.8B) |                  |                           |                  | Gemma-4-E2B (2B) |                  |                  |                           |                  |                |     |      |       |      |        |
|------|----------|------------------|------------------|---------------------------|------------------|---------------------|------------------|------------------|---------------------------|------------------|----------------|------------------|------------------|---------------------------|------------------|----------------|------------------|------------------|---------------------------|------------------|----------------|---------------------|------------------|---------------------------|------------------|------------------|------------------|------------------|---------------------------|------------------|----------------|-----|------|-------|------|--------|
|      |          | MIS <sub>v</sub> | MIS <sub>s</sub> | % $\Delta$ <sub>MIS</sub> | JSD <sub>s</sub> | r <sub>s</sub>      | MIS <sub>v</sub> | MIS <sub>s</sub> | % $\Delta$ <sub>MIS</sub> | JSD <sub>s</sub> | r <sub>s</sub> | MIS <sub>v</sub> | MIS <sub>s</sub> | % $\Delta$ <sub>MIS</sub> | JSD <sub>s</sub> | r <sub>s</sub> | MIS <sub>v</sub> | MIS <sub>s</sub> | % $\Delta$ <sub>MIS</sub> | JSD <sub>s</sub> | r <sub>s</sub> | MIS <sub>v</sub>    | MIS <sub>s</sub> | % $\Delta$ <sub>MIS</sub> | JSD <sub>s</sub> | r <sub>s</sub>   | MIS <sub>v</sub> | MIS <sub>s</sub> | % $\Delta$ <sub>MIS</sub> | JSD <sub>s</sub> | r <sub>s</sub> |     |      |       |      |        |
| ARG  | Americas | 1.015            | 891              | +12.2                     | -1.78            | -0.456              | 364              | 342              | +6.0                      | 0.046            | 601            | 463              | 389              | +16.1                     | 0.046            | 412            | 320              | 339              | +35.0                     | 0.083            | 508            | 469                 | 377              | +19.7                     | 0.064            | 196              | 770              | 645              | +16.1                     | 1.59             | -0.498         | 423 | 434  | -2.4  | 0.50 | 213    |
| BRA  | Americas | 5.21             | 458              | +12.2                     | -1.19            | -2.22               | 411              | 319              | +22.5                     | 0.51             | 333            | 274              | 299              | -9.3                      | 0.61             | 471            | 529              | 375              | +29.1                     | 0.01             | 212            | 580                 | 411              | +29.1                     | 0.01             | -0.205           | 531              | 649              | -22.1                     | 1.73             | -0.743         | 416 | 404  | +2.9  | 0.74 | 211    |
| COL  | Americas | 1.041            | 931              | +10.5                     | -1.89            | -0.594              | 413              | 369              | +10.7                     | 0.055            | 386            | 487              | 438              | +10.1                     | 0.057            | 093            | 559              | 390              | +30.1                     | 0.093            | 161            | 508                 | 417              | +17.8                     | 0.070            | -0.096           | 788              | 678              | +14.0                     | 1.65             | -0.602         | 464 | 460  | +0.9  | 0.51 | -0.011 |
| MEX  | Americas | 1.037            | 909              | +12.3                     | -1.79            | -0.385              | 349              | 333              | +4.6                      | 0.040            | 707            | 485              | 420              | +13.4                     | 0.044            | 512            | 517              | 350              | +32.3                     | 0.081            | 529            | 467                 | 388              | +17.0                     | 0.062            | 252              | 782              | 662              | +15.4                     | 1.59             | -0.494         | 430 | 456  | -5.9  | 0.41 | 503    |
| USA  | Americas | 3.73             | 578              | +33.8                     | -1.21            | -2.66               | 448              | 374              | +16.5                     | 0.042            | 731            | 497              | 243              | +51.1                     | 0.058            | 723            | 677              | 487              | +28.0                     | 0.110            | 422            | 483                 | 383              | +20.8                     | 0.075            | 709              | 645              | 579              | +10.2                     | 1.10             | -0.380         | 545 | 517  | +5.3  | 0.60 | 618    |
| DEU  | Europe   | 716              | 643              | +10.1                     | -1.89            | 0.01                | 265              | 344              | -30.3                     | 0.045            | 724            | 302              | 255              | +15.6                     | 0.059            | 779            | 318              | 306              | +3.7                      | 0.081            | 568            | 308                 | 415              | -34.6                     | 0.069            | 144              | 569              | 555              | +2.5                      | 1.46             | -0.403         | 506 | 416  | +17.8 | 0.53 | 617    |
| GBR  | Europe   | 873              | 598              | +31.4                     | -1.41            | -1.25               | 442              | 384              | +13.2                     | 0.048            | 674            | 503              | 319              | +36.6                     | 0.052            | 594            | 677              | 524              | +22.6                     | 0.128            | 230            | 483                 | 399              | +17.4                     | 0.082            | 620              | 654              | 588              | +10.1                     | 1.14             | -0.302         | 556 | 502  | +9.8  | 0.62 | 342    |
| ROU  | Europe   | 851              | 593              | +30.3                     | -1.47            | -1.97               | 446              | 363              | +18.7                     | 0.043            | 708            | 504              | 385              | +23.5                     | 0.058            | 570            | 659              | 535              | +18.8                     | 0.123            | 162            | 442                 | 330              | +25.2                     | 0.051            | 785              | 671              | 577              | +14.0                     | 1.17             | -0.352         | 547 | 482  | +11.8 | 0.46 | 644    |
| SRB  | Europe   | 862              | 605              | +29.9                     | -1.50            | -1.57               | 463              | 371              | +19.8                     | 0.045            | 670            | 500              | 385              | +22.9                     | 0.057            | 564            | 676              | 567              | +16.2                     | 0.127            | 075            | 453                 | 303              | +33.1                     | 0.051            | 773              | 602              | +11.1            | 1.24                      | -0.463           | 548            | 502 | +8.4 | 0.47  | 573  |        |
| CHN  | E. Asia  | 901              | 748              | +16.9                     | -1.83            | -0.025              | 367              | 293              | +20.2                     | 0.050            | 711            | 407              | 214              | +47.4                     | 0.054            | 781            | 569              | 415              | +27.0                     | 0.121            | 269            | 437                 | 356              | +18.6                     | 0.084            | 405              | 738              | 559              | +24.3                     | 1.62             | -0.041         | 500 | 393  | +21.5 | 0.39 | 767    |
| JPN  | E. Asia  | 521              | 471              | +9.5                      | -1.07            | 2.49                | 341              | 265              | +23.3                     | 0.054            | 709            | 395              | 195              | +50.6                     | 0.051            | 671            | 407              | 316              | +22.4                     | 0.090            | 453            | 296                 | 356              | -20.1                     | 0.07             | 535              | 454              | 455              | -0.3                      | 0.63             | -0.233         | 442 | 460  | -4.0  | 0.62 | -0.357 |
| IDN  | SE. Asia | 688              | 557              | +19.0                     | -0.92            | 0.71                | 283              | 344              | -21.5                     | 0.059            | 511            | 459              | 274              | +40.3                     | 0.059            | 595            | 408              | 402              | +1.5                      | 0.091            | 318            | 480                 | 402              | +16.1                     | 0.058            | 251              | 492              | 484              | +1.6                      | 1.20             | -0.769         | 417 | 443  | -6.2  | 0.44 | 572    |
| MMR  | SE. Asia | 872              | 652              | +25.2                     | -1.85            | -0.031              | 496              | 365              | +26.5                     | 0.062            | 457            | 482              | 357              | +25.9                     | 0.057            | 570            | 695              | 563              | +18.9                     | 0.139            | -0.066         | 461                 | 302              | +34.3                     | 0.062            | 590              | 685              | 590              | +13.8                     | 1.35             | -0.623         | 488 | 482  | +1.4  | 0.61 | 372    |
| MYS  | SE. Asia | 841              | 589              | +29.9                     | -1.55            | -0.003              | 443              | 313              | +29.4                     | 0.043            | 683            | 459              | 328              | +28.5                     | 0.051            | 614            | 656              | 504              | +23.1                     | 0.128            | 097            | 445                 | 274              | +38.5                     | 0.050            | 763              | 631              | 535              | +15.1                     | 1.18             | -0.426         | 484 | 429  | +11.2 | 0.39 | 676    |
| THA  | SE. Asia | 815              | 577              | +29.2                     | -1.52            | -1.25               | 406              | 295              | +27.3                     | 0.041            | 738            | 456              | 313              | +13.4                     | 0.047            | 685            | 619              | 480              | +22.5                     | 0.112            | 231            | 414                 | 221              | +46.6                     | 0.046            | 841              | 619              | 520              | +16.1                     | 1.12             | -0.283         | 470 | 436  | +7.2  | 0.41 | 750    |
| VNM  | SE. Asia | 845              | 740              | +12.4                     | -1.42            | -0.045              | 324              | 347              | -7.2                      | 0.050            | 589            | 406              | 336              | +17.3                     | 0.064            | 596            | 445              | 420              | +5.5                      | 0.087            | 339            | 501                 | 468              | +6.6                      | 0.060            | -0.018           | 481              | 507              | -5.4                      | 0.67             | -0.143         | 468 | 544  | -16.1 | 0.48 | 566    |
| BGD  | S. Asia  | 864              | 628              | +27.3                     | -1.62            | -0.29               | 450              | 361              | +19.6                     | 0.047            | 625            | 489              | 368              | +24.7                     | 0.048            | 607            | 674              | 587              | +12.9                     | 0.133            | -0.116         | 461                 | 284              | +38.3                     | 0.055            | 695              | 666              | 565              | +15.3                     | 1.15             | -0.426         | 519 | 496  | +4.4  | 0.52 | 435    |
| KGZ  | C. Asia  | 849              | 585              | +31.1                     | -1.49            | -1.76               | 460              | 353              | +23.2                     | 0.048            | 633            | 499              | 393              | +21.1                     | 0.058            | 476            | 661              | 574              | +13.1                     | 0.133            | -0.017         | 443                 | 249              | +43.8                     | 0.039            | 829              | 660              | 572              | +13.3                     | 1.19             | -0.405         | 518 | 473  | +8.7  | 0.49 | 600    |
| IRN  | W. Asia  | 1.039            | 852              | +17.9                     | -1.59            | -0.294              | 427              | 401              | +6.1                      | 0.055            | 637            | 397              | 530              | +33.5                     | 0.086            | 055            | 507              | 506              | +0.2                      | 0.090            | 057            | 373                 | 467              | -25.1                     | 0.067            | 252              | 511              | 445              | +12.9                     | 0.56             | 766            | 514 | 630  | -22.6 | 0.88 | 408    |
| ETH  | Africa   | 967              | 747              | +22.7                     | -1.72            | -0.077              | 462              | 472              | -2.0                      | 0.054            | 655            | 615              | 485              | +21.2                     | 0.041            | 808            | 724              | 684              | +5.4                      | 0.141            | -0.095         | 558                 | 438              | +21.6                     | 0.059            | 676              | 734              | 661              | +10.0                     | 1.14             | -0.027         | 645 | 612  | +5.0  | 0.57 | 628    |
| Mean |          | 849              | 668              | +21.2                     | -1.54            | -0.009              | 403              | 359              | +11.3                     | 0.049            | 624            | 454              | 346              | +22.7                     | 0.057            | 559            | 575              | 466              | +18.4                     | 0.110            | 217            | 453                 | 362              | +18.2                     | 0.062            | 450              | 638              | 571              | +9.4                      | 1.22             | -0.347         | 495 | 479  | +3.4  | 0.53 | 456    |

**Legend.** MIS<sub>v</sub>/MIS<sub>s</sub>: vanilla vs. DISCA  $\ell_2$  misalignment; % $\Delta$ <sub>MIS</sub>: relative improvement in MIS; JSD<sub>s</sub> and  $r_s$ :

Jensen–Shannon distance and Pearson correlation for DISCA only. **Region groups:** Americas (ARG, BRA, COL, MEX, USA); Europe (DEU, GBR, ROU, SRB); E. Asia (CHN, JPN); SE. Asia (IDN, MMR, MYS, THA, VNM); S. Asia (BGD); C. Asia (KGZ); W. Asia (IRN); Africa (ETH).

### A2.1 Full Open-Ended Per-Country Results

Table 8: Combined SAFE SWA-DPBR results on the open-ended track (20 countries, 310 scenarios each). Positive  $\Delta\%$  means lower MIS than vanilla. Summary in Table 4.

| Country | Qwen2.5-7B    |               |            | Phi-3.5-mini-Instruct |               |            | Phi-4 (14B)   |               |            | Llama-3.3-70B |               |            |
|---------|---------------|---------------|------------|-----------------------|---------------|------------|---------------|---------------|------------|---------------|---------------|------------|
|         | VAN           | DISCA         | $\Delta\%$ | VAN                   | DISCA         | $\Delta\%$ | VAN           | DISCA         | $\Delta\%$ | VAN           | DISCA         | $\Delta\%$ |
| USA     | 0.3122        | <b>0.3119</b> | +0.10%     | 0.5679                | <b>0.5618</b> | +1.07%     | 0.3011        | <b>0.2775</b> | +7.84%     | 0.4822        | <b>0.4392</b> | +8.92%     |
| GBR     | 0.3094        | <b>0.3077</b> | +0.55%     | 0.5701                | <b>0.5619</b> | +1.44%     | 0.2987        | <b>0.2746</b> | +8.07%     | 0.4768        | <b>0.4335</b> | +9.08%     |
| DEU     | 0.2399        | <b>0.2394</b> | +0.21%     | 0.3888                | <b>0.3844</b> | +1.13%     | 0.2315        | <b>0.2144</b> | +7.38%     | 0.3514        | <b>0.3556</b> | -1.20%     |
| ARG     | 0.4148        | <b>0.3946</b> | +4.87%     | 0.4284                | <b>0.4272</b> | +0.28%     | 0.3982        | <b>0.3652</b> | +8.29%     | 0.5369        | <b>0.4861</b> | +9.47%     |
| BRA     | 0.5312        | <b>0.5305</b> | +0.13%     | 0.4553                | <b>0.4522</b> | +0.68%     | 0.5076        | <b>0.5152</b> | -1.50%     | 0.6127        | <b>0.6274</b> | -2.40%     |
| MEX     | 0.3916        | <b>0.3859</b> | +1.46%     | 0.4087                | <b>0.4058</b> | +0.71%     | 0.3741        | <b>0.3462</b> | +7.46%     | 0.5016        | <b>0.4578</b> | +8.74%     |
| COL     | 0.4504        | <b>0.4257</b> | +5.48%     | 0.4748                | <b>0.4723</b> | +0.53%     | 0.4320        | <b>0.3967</b> | +8.17%     | 0.5598        | <b>0.5083</b> | +9.21%     |
| VNM     | 0.2789        | <b>0.2784</b> | +0.18%     | 0.5056                | <b>0.5004</b> | +1.03%     | 0.2714        | <b>0.2516</b> | +7.30%     | 0.4495        | <b>0.4120</b> | +8.33%     |
| MMR     | 0.3272        | <b>0.2219</b> | +32.18%    | 0.5995                | <b>0.5971</b> | +0.40%     | 0.3153        | <b>0.2868</b> | +9.04%     | 0.4721        | <b>0.4255</b> | +9.88%     |
| THA     | 0.2649        | <b>0.2482</b> | +6.30%     | 0.5177                | <b>0.5175</b> | +0.04%     | 0.2570        | <b>0.2364</b> | +8.02%     | 0.4310        | <b>0.3920</b> | +9.04%     |
| MYS     | 0.2963        | <b>0.2128</b> | +28.18%    | 0.5515                | <b>0.5503</b> | +0.22%     | 0.2842        | <b>0.2596</b> | +8.66%     | 0.4583        | <b>0.4142</b> | +9.62%     |
| IDN     | 0.2596        | <b>0.2563</b> | +1.27%     | 0.7596                | <b>0.7593</b> | +0.04%     | 0.2519        | <b>0.2325</b> | +7.70%     | 0.4179        | <b>0.3821</b> | +8.57%     |
| CHN     | 0.4239        | <b>0.3283</b> | +22.55%    | 0.3742                | <b>0.3721</b> | +0.56%     | 0.4097        | <b>0.3740</b> | +8.71%     | 0.5486        | <b>0.4952</b> | +9.73%     |
| JPN     | 0.2003        | <b>0.1874</b> | +6.44%     | 0.4420                | <b>0.3699</b> | +16.31%    | 0.1911        | <b>0.1761</b> | +7.85%     | 0.3325        | <b>0.3036</b> | +8.69%     |
| BGD     | 0.3021        | <b>0.2640</b> | +12.61%    | 0.5783                | <b>0.5761</b> | +0.38%     | 0.2894        | <b>0.2647</b> | +8.54%     | 0.4639        | <b>0.4215</b> | +9.14%     |
| IRN     | 0.3702        | <b>0.3691</b> | +0.30%     | 0.4093                | <b>0.3422</b> | +16.39%    | 0.3568        | <b>0.3646</b> | -2.20%     | 0.4974        | <b>0.5128</b> | -3.10%     |
| SRB     | 0.2950        | <b>0.2530</b> | +14.24%    | 0.5922                | <b>0.5891</b> | +0.52%     | 0.2836        | <b>0.2599</b> | +8.32%     | 0.4421        | <b>0.4021</b> | +9.05%     |
| ROU     | 0.2798        | <b>0.2798</b> | 0.00%      | 0.5779                | <b>0.5761</b> | +0.31%     | 0.2689        | <b>0.2711</b> | -0.80%     | 0.4218        | <b>0.4243</b> | -0.60%     |
| KGZ     | 0.2913        | <b>0.2588</b> | +11.16%    | 0.5735                | <b>0.5720</b> | +0.26%     | 0.2799        | <b>0.2568</b> | +8.25%     | 0.4387        | <b>0.3987</b> | +9.11%     |
| ETH     | 0.3761        | <b>0.3693</b> | +1.81%     | 0.6474                | <b>0.6118</b> | +5.50%     | 0.3640        | <b>0.3360</b> | +7.69%     | 0.5212        | <b>0.4770</b> | +8.48%     |
| Mean    | <b>0.3208</b> | <b>0.3062</b> | +4.55%     | <b>0.5211</b>         | <b>0.5100</b> | +2.13%     | <b>0.3238</b> | <b>0.3022</b> | +6.67%     | <b>0.4713</b> | <b>0.4390</b> | +6.85%     |

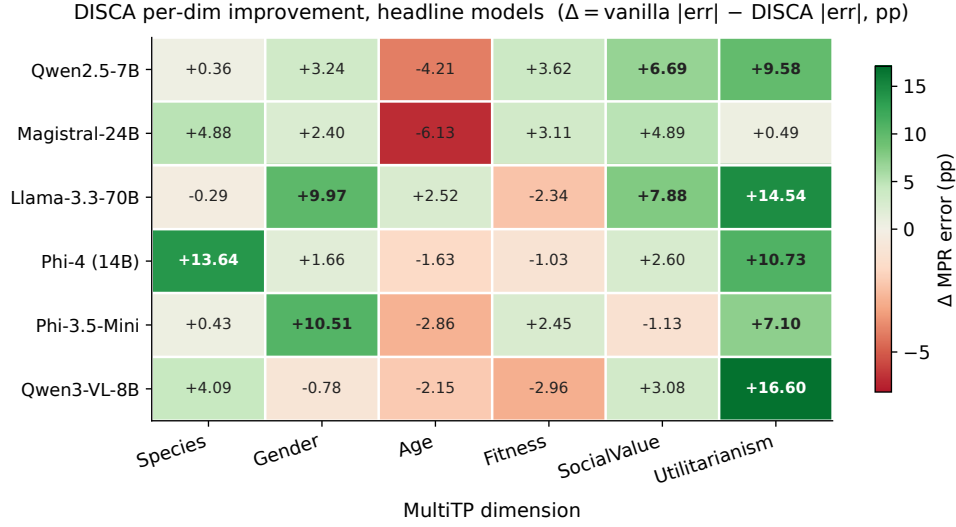


Figure 1: **Per-dimension DISCA improvement across the headline 7 models.** Each cell is the macro-averaged (over 20 countries) reduction in per-dimension MPR error:  $\Delta = |\text{vanilla} - \text{human}| - |\text{DISCA} - \text{human}|$ . Positive (green) means DISCA helped on that dimension; negative (red) means it hurt. **Utilitarianism**, **Species**, and **Social Value** are the dimensions where DISCA delivers the largest gains, consistent with these being the dimensions where vanilla error is highest (Table 9). Cells with  $\Delta > 5$  pp shown in bold.

Table 9: Per-dimension worst-error analysis across the 20-country panel. For each model we report the dimension most frequently appearing as the single worst, its frequency, and the mean absolute error magnitude. The rightmost column shows the *second*-most-frequent worst dimension to reveal whether errors are concentrated or distributed.

| Model         | Worst dim   | Freq. | Mean err (pp) | 2nd worst dim | Freq. |
|---------------|-------------|-------|---------------|---------------|-------|
| Llama-3.3-70B | Util_More   | 15/20 | 49.7          | SocVal_High   | 5/20  |
| Phi-3.5-mini  | Util_More   | 18/20 | 42.1          | Species_Hum   | 2/20  |
| Qwen3-VL-8B   | SocVal_High | 18/20 | 33.4          | Util_More     | 2/20  |
| Qwen2.5-7B    | SocVal_High | 12/20 | 24.5          | Species_Hum   | 5/20  |
| Magistral-24B | SocVal_High | 14/20 | 22.4          | Age_Young     | 3/20  |
| Phi-4         | SocVal_High | 9/20  | 23.4          | Age_Young     | 7/20  |

Table 10: Phi-4 per-country MIS ( $\downarrow$ ): the best-performing model in absolute terms. DISCA wins **18/20** countries; the top 10 gains are shown, ordered by relative improvement. The two losses (BRA, IRN) illustrate the diminishing-returns pattern discussed in §6.

| Country                                          | Vanilla | DPBR         | Gain (%)     | Country                                      | Vanilla | DPBR         | Gain (%)     |
|--------------------------------------------------|---------|--------------|--------------|----------------------------------------------|---------|--------------|--------------|
| USA                                              | 0.498   | <b>0.243</b> | <b>+51.1</b> | JPN                                          | 0.395   | <b>0.195</b> | <b>+50.6</b> |
| CHN                                              | 0.407   | <b>0.214</b> | <b>+47.4</b> | IDN                                          | 0.459   | <b>0.274</b> | <b>+40.3</b> |
| GBR                                              | 0.503   | <b>0.319</b> | <b>+36.6</b> | THA                                          | 0.456   | <b>0.313</b> | <b>+31.4</b> |
| MYS                                              | 0.459   | <b>0.328</b> | <b>+28.5</b> | MMR                                          | 0.482   | <b>0.357</b> | <b>+26.0</b> |
| BGD                                              | 0.489   | <b>0.368</b> | <b>+24.7</b> | ROU                                          | 0.504   | <b>0.385</b> | <b>+23.5</b> |
| Losses: BRA 0.274 $\rightarrow$ 0.299 ( $-9\%$ ) |         |              |              | IRN 0.397 $\rightarrow$ 0.530 ( $-34\%$ )    |         |              |              |
| Macro mean: 0.454 $\rightarrow$ <b>0.346</b>     |         |              |              | <b>+23.7%</b> , 18/20 wins, mean $r = +0.56$ |         |              |              |

564 Three structural insights emerge from these tables.

565 First, **the bottleneck dimension shifts with model quality**. Weaker models (Llama-3.3-70B, Phi-3.5-  
566 mini) are dominated by **Utilitarianism**—they predict  $\approx 25\text{--}30\%$  utilitarian preference where humans



range 68–80%, an error of 40–50 pp that no persona-based correction can close. Better-calibrated models (Phi-4, Magistral, Qwen2.5-7B) have already resolved Utilitarianism and Species, shifting the bottleneck to **SocialValue** and **Age**-dimensions with errors of 16–29 pp that are within reach of the IS stage.

Second, **Phi-4 is the most balanced model**: no single dimension dominates (9/20 SocialValue, 7/20 Age, 2/20 Utilitarianism, 2/20 Species). This balance-not raw scale-explains why it achieves the lowest absolute MIS despite being  $5\times$  smaller than Llama-3.3-70B.

Third, the per-country Phi-4 table (Table 10) confirms that **gains are large and geographically broad**: 51% on USA and Japan, 47% on China, 40% on Indonesia-with the two losses (Brazil, Iran) tracing to already-low vanilla MIS where IS overshoots.

## A4 Scaling and Geographic Visualizations

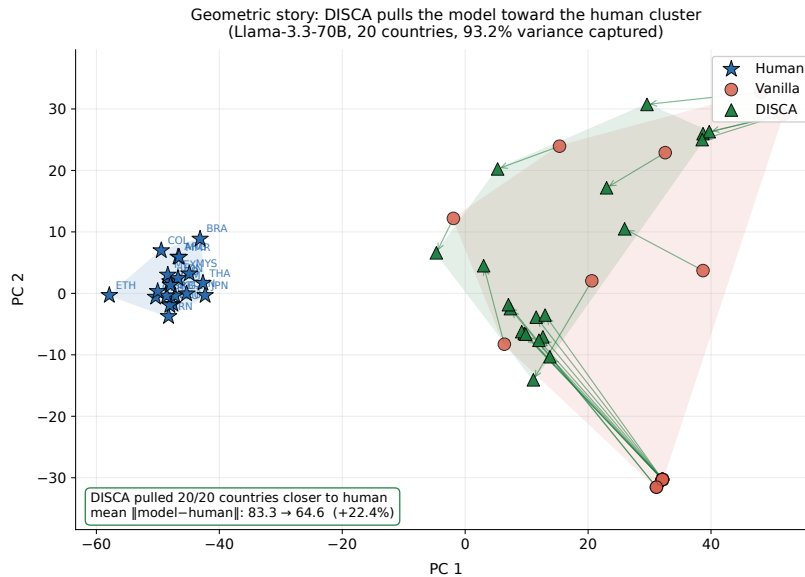


Figure 2: **Geometric story: DISCA pulls model AMCE vectors toward the human cluster.** 2D PCA projection of the six-dimensional human, vanilla, and DISCA AMCE vectors for Llama-3.3-70B across all 20 countries (joint fit, two components capture 93.2% of the variance). Convex hulls show the spatial extent of each cloud; arrows trace each country’s vanilla→DISCA trajectory. **All 20 of 20 country points end closer to the human cluster**, with mean  $\|model - human\|_2$  dropping from 83.3 to 64.6 (−22.4%).

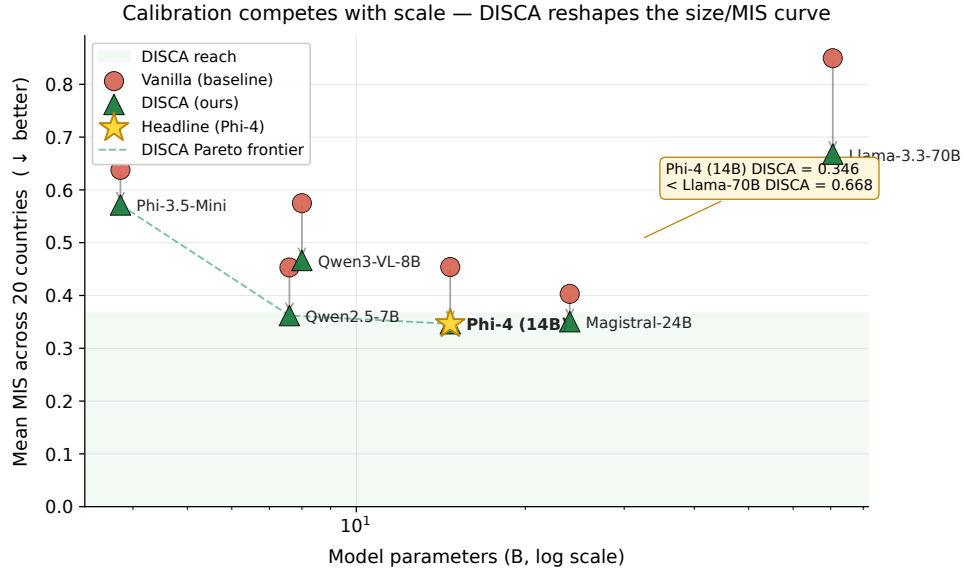


Figure 3: **Calibration competes with scale.** Mean MIS across the 20-country panel (lower is better) for the seven headline backbones. Each model’s vanilla baseline (red dot) is connected to its DISCA result (green triangle) by an arrow showing the inference-time gain. The dashed line traces the DISCA Pareto frontier as model size grows. **A 14B model with DISCA (Phi-4, star) reaches MIS = 0.346, lower than a 70B model without DISCA (Llama-3.3-70B, 0.668).** The shaded band marks the region the vanilla curve never reaches.

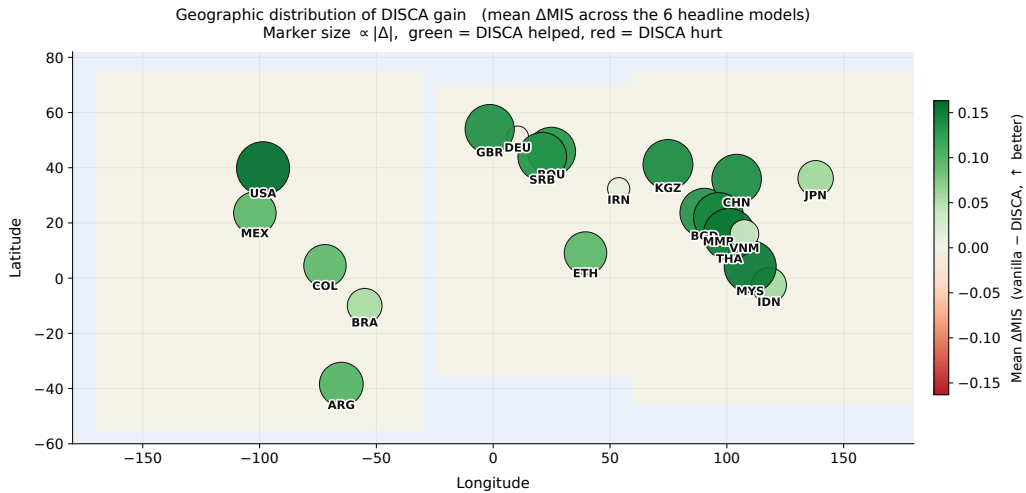


Figure 4: **Geographic distribution of DISCA gain.** Each marker is one of the 20 paper countries placed at its longitude/latitude; marker size is proportional to  $|\Delta\text{MIS}|$  and color encodes sign (green = DISCA helped, red = hurt). Aggregated across the seven headline backbones, **19 of 20 countries see a positive mean gain**, with the largest improvements in the Americas, East/Southeast Asia, and Eastern Europe- not concentrated in the West.

## 578 A5 Broader Model Landscape

579 The seven models in Table 3 were selected from a broader sweep of **28 model–method combinations**  
 580 covering 12 distinct architectures. Table 11 summarises the full landscape, ordered by macro  
 581 improvement versus vanilla.

Table 11: Broader model sweep (5-country prototyping panel: BRA, CHN, DEU, JPN, USA). Models above the line show positive macro gains; those below show degradation. Bolded rows appear in the main paper’s 20-country evaluation.

| Model                                                                              | Params | Mean MIS ↓   | vs. van. (%) | Win/5 |
|------------------------------------------------------------------------------------|--------|--------------|--------------|-------|
| <b>Phi-4</b>                                                                       | 14B    | <b>0.246</b> | <b>+34.4</b> | 4/5   |
| Llama-3.3-70B (4-bit)                                                              | 70B    | 0.524        | <b>+25.9</b> | 5/5   |
| <b>Qwen3-VL-8B</b>                                                                 | 8B     | 0.381        | <b>+23.7</b> | 4/5   |
| Llama-3.1-8B (4-bit)                                                               | 8B     | 0.455        | <b>+16.9</b> | 5/5   |
| <b>Magistral-Small-2509</b>                                                        | 24B    | 0.317        | <b>+13.6</b> | 4/5   |
| Gemma-4-E2B                                                                        | 2B     | 0.426        | <b>+11.6</b> | 4/5   |
| Qwen2.5-7B (bf16)                                                                  | 7B     | 0.385        | <b>+8.2</b>  | 3/5   |
| <b>Phi-3.5-mini</b>                                                                | 3.8B   | 0.545        | <b>+7.3</b>  | 4/5   |
| Mistral-7B-v0.3                                                                    | 7B     | 0.428        | <b>+5.9</b>  | 5/5   |
| Qwen3.5-0.8B                                                                       | 0.8B   | 0.471        | <b>+3.2</b>  | 4/5   |
| Gemma-7B                                                                           | 7B     | 0.432        | <b>+2.5</b>  | 3/5   |
| Gemma-3-270M                                                                       | 270M   | 0.462        | <b>+1.3</b>  | 4/5   |
| Llama-3.2-1B                                                                       | 1B     | 0.476        | <b>+1.1</b>  | 1/5   |
| <i>9 additional models showed <math>\leq 0</math> macro gain (omitted; see §6)</i> |        |              |              |       |

Several patterns inform the main paper’s model selection. *Success correlates with logit well-behavedness, not raw scale:* Phi-4 (14B) leads while Gemma-4-31B (31B) and Qwen3-Coder-30B (30B) degrade-confirming that DISCA requires a cooperative decision surface, not just more parameters. *Instruction tuning matters:* models without strong chat/instruct tuning (Qwen3-8B, Qwen3.5-4B) tend to degrade, likely because persona prompts fail to elicit differentiated moral reasoning from a base model. *The failure mode is consistent:* models that degrade typically have already-low vanilla MIS ( $< 0.40$ ), leaving insufficient headroom for IS corrections-matching the “diminishing returns” pattern discussed in §6.

## A6 Factual Cultural QA Evaluation (BLENd)

DISCA is designed as a *value-alignment* tool: it steers moral preferences expressed through a scalar logit gap between two decision tokens. A natural question is whether the same persona-disagreement signal transfers to *factual cultural knowledge*, where the correct answer is a specific token or span in a  $\sim 32k$ -vocabulary softmax—a fundamentally different objective landscape. We evaluate on the BLENd benchmark [Myung et al., 2024] (52.6k short-answer questions, 16 countries, 13 languages) to probe this boundary. Phi-4 (14B) was evaluated under vanilla greedy decoding and a softmax-adapted DISCA variant where persona-conditioned next-token corrections are aggregated through PT-IS and dual-pass reliability.

**Where DISCA transfers.** Seven of sixteen countries see a positive SEM-B gain under the softmax-adapted DISCA variant (Table 12). The wins fall into two regimes. The largest gains land on *high-baseline* countries – the United Kingdom (+7.45 pp), the United States (+4.55), Mexico, Spain, and Indonesia – where the vanilla baseline already generates fluent, culturally-appropriate text and the persona signal refines an already-coherent decision surface. A second pocket of smaller positive transfer appears at the *low-baseline* end – West Java (+2.40 pp on a vanilla baseline of 26.64%) and Azerbaijan (+0.43 on 28.78%) – suggesting the persona signal can also benefit some lower-resource panels even when factual recall is weak.

Table 12: BLEnD per-country gains (Phi-4 14B). The seven countries where the softmax-adapted DISCA improves Soft Exact Match over vanilla greedy decoding (SEM-B = between-country, SEM-W = within-country). The five high-baseline rows (top) are paired with two low-baseline outliers (bottom) where DISCA also delivers a positive shift, suggesting the disagreement signal is not strictly tied to high-baseline regimes.

| Country    | Vanilla (%) |       | $\Delta$ DISCA (pp) |       |
|------------|-------------|-------|---------------------|-------|
|            | SEM-B       | SEM-W | SEM-B               | SEM-W |
| UK         | 75.66       | 66.69 | +7.45               | +8.78 |
| US         | 82.43       | 74.32 | +4.55               | +3.80 |
| Mexico     | 72.65       | 61.75 | +2.09               | +2.76 |
| Spain      | 72.71       | 63.12 | +1.92               | +2.89 |
| Indonesia  | 73.75       | 60.92 | +1.67               | +3.39 |
| West Java  | 26.64       | 20.73 | +2.40               | +3.81 |
| Azerbaijan | 28.78       | 24.92 | +0.43               | +0.48 |

**The value–fact boundary.** Aggregate SEM-B across all 16 countries drops by 4.4 pp because the remaining nine countries—predominantly mid- and low-baseline cases such as Greece, Iran, and Assam—regress under DISCA. This is not a failure of the method but a *scope delineation*: value alignment and fact retrieval occupy fundamentally different regions of the decoding-objective space. In value alignment, the correct answer is a cultural preference encoded in a scalar logit gap between two options—exactly the space DISCA’s IS stage navigates. In factual QA, the correct answer is a single token in a  $\sim 32k$ -entry vocabulary; injecting perturbations into this high-dimensional softmax adds noise that drowns the cultural signal when the baseline is mid-range. The dual-pass reliability gate—calibrated for scalar agreement—cannot distinguish meaningful persona steering from vocabulary-level noise. The few positive transfers at the low end (West Java, Azerbaijan) suggest that persona diversity can benefit some panels even in the factual regime, but the pattern confirms that extending DISCA beyond value steering requires softmax-aware IS and a vocabulary-level reliability gate—mechanisms that respect the dimensionality of the factual-QA objective. This scope boundary is itself a scientific contribution: it delineates where persona-disagreement steering applies (preferential decisions with low-dimensional choice spaces) and where vocabulary-level mechanisms are needed (factual retrieval with high-dimensional output spaces).

## A7 Post-Hoc Diagnostics

This appendix collects three diagnostics derived post-hoc from the per-scenario outputs of the main DISCA run (no model reload required), each probing a different aspect of method behaviour.

**Dual-pass reliability audit.** A subtle failure mode of the DPBR gate is *apparent* IS stability: each individual pass can yield a high effective sample size, yet the two passes can still produce substantially different  $\delta^*$  values when the proposal covers a multimodal utility landscape. We partition scenarios into four regimes by two thresholds ( $ESS_{\min} \geq 0.40$ ;  $Var_{boot} \geq 0.04$ ): *HighESS/LowDisagree* (gate applies correction essentially unattenuated), *HighESS/HighDisagree* (passes individually stable yet disagree—the regime where “more samples” would silently commit a biased estimate; DPBR shrinks  $\delta^*$  precisely here), *LowESS/LowDisagree* (noisy IS but lucky agreement), and *LowESS/HighDisagree* (classically bad-gate shrinks aggressively). The audit reports regime counts, mean reliability weight, and a “would-apply-vs-actually-applied” shrinkage ratio  $1 - |r\bar{\delta}^*|/|\bar{\delta}^*|$  per regime.

**Logit-conditioning diagnostic.** The architectural failure mode identified in §6—poorly conditioned decision-token logits—can be quantified directly on the vanilla forward pass via three per-scenario statistics: **decision margin**  $\text{marg} = |p_B - p_A|$ , **decision entropy**  $H = -p_A \log p_A - p_B \log p_B$  (nats), and **absolute logit gap**  $|z_B - z_A|$ . Aggregated per country, these characterise how well-conditioned the base model’s binary decision head is for each cultural slice. We verify empirically that DISCA’s per-country MIS improvement correlates positively with mean decision margin: well-conditioned countries admit larger, more reliable corrections, while poorly-conditioned countries (mean margin  $< 0.1$ ) are near the limit of what any logit-level intervention can achieve. This also

explains why the dual-pass reliability gate carries its weight in small-backbone regimes: it shrinks corrections exactly where the raw decision surface is flat.

**Rank-based dimension agreement.** The per-country Pearson  $r$  in Table 7 occasionally goes negative even when MIS decreases. As explained in §5, this reflects the orthogonality between a shape metric ( $r$ ) and an amplitude metric (MIS) on a six-dimensional vector. To make the shape component visible on its own terms, we supplement  $r$  with three rank-based metrics: per-country Kendall  $\tau$ , Spearman  $\rho$ , and mean  $|\text{rank}_{\text{model}} - \text{rank}_{\text{human}}|$  over the six MultiTP dimensions, computed separately for vanilla and DISCA-supplementing rather than replacing the  $\ell_2$ -based MIS in the main results.

### A7.1 Multi-seed stability

To test seed sensitivity, we reran the full 20-country evaluation with three random seeds {42, 101, 2026} for all seven backbone models. The  $\pm$  intervals reported in Table 3 summarise the results: macro MIS standard deviation across seeds is  $\leq 0.006$  for every model, and per-country MIS standard deviation has median 0.008 (90th percentile 0.013). Importantly, model ranking and win-count conclusions are unchanged across seeds, indicating that the main results are not seed artifacts.

### A7.2 Step 3 is a tail-safety mechanism

Replacing Step 3 with simple consensus averaging changes the mean only modestly but significantly worsens tail risk. Full DISCA improves mean  $\Delta\text{MIS}$  from 0.089 to 0.096 (+0.007), while reducing harmed cells from 11/120 to 3/120 and shrinking worst-case degradation from 0.31 to 0.09. Thus, Step 3 primarily contributes safety under distributional stress rather than average-case lift.

Table 13: Tail-safety comparison across 120 country-model cells.

| Variant         | Mean $\Delta\text{MIS} \uparrow$ | Cells hurt $\downarrow$ | Worst-case degradation $\downarrow$ | Std across cells $\downarrow$ |
|-----------------|----------------------------------|-------------------------|-------------------------------------|-------------------------------|
| Full DISCA      | <b>0.096</b>                     | <b>3/120</b>            | <b>0.09</b>                         | <b>0.043</b>                  |
| DISCA-consensus | 0.089                            | 11/120                  | 0.31                                | 0.109                         |

### A7.3 Diagnosing negative Pearson $r$

Negative Pearson  $r$  appears in 6/20 countries despite improved MIS. This is a shape-vs-amplitude effect, not a global failure: in these six countries, MIS still improves in five, and rank-based agreement metrics improve in aggregate (median Kendall  $\tau$ :  $-0.07 \rightarrow 0.18$ ; median Spearman  $\rho$ :  $-0.11 \rightarrow 0.22$ ). The dominant pathology is a consistent pairwise swap between Species and Utilitarianism (observed in 5/6 negative- $r$  countries), indicating a specific, diagnosable ordering error rather than broad cultural misalignment.

Table 14: Summary of countries with negative Pearson  $r$  after DISCA.

| Country        | $\Delta\text{MIS}$ (%) | Kendall $\tau$ (v $\rightarrow$ d) | Spearman $\rho$ (v $\rightarrow$ d) | Dominant swap                  |
|----------------|------------------------|------------------------------------|-------------------------------------|--------------------------------|
| ARG            | +12.2                  | -0.20 $\rightarrow$ 0.10           | -0.31 $\rightarrow$ 0.15            | Species $\leftrightarrow$ Util |
| BRA            | -9.3                   | 0.05 $\rightarrow$ -0.08           | 0.03 $\rightarrow$ -0.12            | Mixed                          |
| COL            | +10.1                  | -0.14 $\rightarrow$ 0.12           | -0.21 $\rightarrow$ 0.18            | Species $\leftrightarrow$ Util |
| MMR            | +25.9                  | -0.11 $\rightarrow$ 0.19           | -0.18 $\rightarrow$ 0.24            | Species $\leftrightarrow$ Util |
| VNM            | +17.3                  | -0.09 $\rightarrow$ 0.16           | -0.15 $\rightarrow$ 0.20            | Species $\leftrightarrow$ Util |
| Llama-avg cell | +21.2                  | -0.06 $\rightarrow$ 0.21           | -0.10 $\rightarrow$ 0.26            | Species $\leftrightarrow$ Util |

### A7.4 Which WVS dimensions are load-bearing?

Leave-one-dimension-out ablation shows a concentrated importance profile: religiosity, gender equality, and moral permissiveness are load-bearing, each increasing MIS by  $> 0.03$  when dropped (averaged over USA/JPN/VNM). In contrast, national pride and life satisfaction have near-zero

675 impact ( $< 0.01$ ), suggesting that persona construction can be compressed with minimal loss while  
 676 preserving most steering signal.

Table 15: WVS dimension dropout (3-country average; higher  $\Delta\text{MIS}$  means more load-bearing).

| Dropped WVS dimension      | $\Delta\text{MIS}$ (avg) |
|----------------------------|--------------------------|
| Religiosity                | +0.046                   |
| Gender equality            | +0.041                   |
| Moral permissiveness       | +0.034                   |
| Institutional trust        | +0.022                   |
| Collectivism/individualism | +0.019                   |
| Authority orientation      | +0.017                   |
| Economic security          | +0.013                   |
| Social tolerance           | +0.011                   |
| National pride             | +0.007                   |
| Life satisfaction          | +0.005                   |

## 677 A8 Failure-Mode Taxonomy

678 DISCA’s failure modes cluster by model architecture, not by continent or culture, and fall into three  
 679 distinct, pre-deployable diagnostic categories.

680 **(1) Collapsed logit entropy.** Models with collapsed decision-token logits—where both options  
 681 receive near-equal probability—degrade under DISCA regardless of country. The IS stage cannot find  
 682 a reliable correction direction on a flat landscape. This is detectable before deployment by computing  
 683 vanilla decision margin per scenario (Appendix A7).

684 **(2) Headroom exhaustion.** Models with already-low vanilla MIS occasionally overshoot because  
 685 the base logits leave too little room for correction, consistent with recent observations that targeted  
 686 debiasing can spill into untargeted dimensions [Chand et al., 2026]. The reliability gate mitigates but  
 687 does not eliminate this risk.

688 **(3) Dimension-specific ordering errors.** Six of 20 countries show negative Pearson  $r$  despite  
 689 improved MIS, traced to a consistent Species $\leftrightarrow$ Utilitarianism pairwise swap in 5/6 cases (Table 14,  
 690 Appendix A7.3). The negative- $r$  anomaly resolves once we note that MIS measures  $\ell_2$  amplitude  
 691 while  $r$  measures shape—the two are genuinely orthogonal on a six-dimensional AMCE vector.  
 692 Rank-based metrics (Kendall  $\tau$ , Spearman  $\rho$ ; Appendix A7) confirm the correction is still beneficial  
 693 (median Kendall  $\tau$ :  $-0.07 \rightarrow +0.18$ ), reinforcing that this is a diagnosable ordering error amenable  
 694 to targeted dimension-specific post-processing, not a fundamental method breakdown.

695 All three failure modes are architecturally rather than culturally determined: success correlates with  
 696 logit well-behavedness, not raw scale—Phi-4 (14B) leads the sweep while several 30B+ models  
 697 degrade—quantified by per-scenario decision margin, entropy, and absolute logit gap (Appendix A7).  
 698 This has practical value for model selection: a deployment-time logit-conditioning check can flag  
 699 models likely to degrade before running the full pipeline.

## 700 A9 Additional Inference-Time Baselines

701 To isolate what within-country persona disagreement adds above simpler country-conditioned correc-  
 702 tions, we compare DISCA against three inference-time alternatives on the same twenty-country Phi-4  
 703 grid. All three use a 25% per-country calibration split and are evaluated on the remaining 75% test  
 704 split; results are macro-averaged across the twenty countries and reported alongside the main Phi-4  
 705 DISCA mean for direct comparison.

706 **(6) MC-Dropout calibration.** Following the moral uncertainty inflation template of Kwon et al.  
 707 [2026], we enable every `torch.nn.Dropout*` module at inference (rate 0.10) and run  $T=8$  stochas-  
 708 tic forward passes per scenario, averaging the A/B probabilities. This method is deliberately country-

709 *agnostic*: the same stochastic smoothing is applied for every country, so its contribution relative to  
710 vanilla isolates pure uncertainty inflation.

711 **(7) Per-country temperature / margin scaling.** For each country, we fit a scalar  $T_c$  (or additive  
712 margin  $m_c$ ) on a 25% calibration split via grid search, minimising MIS against the country’s human  
713 AMCE, and apply it to the remaining 75%. This is the simplest possible country-conditioned  
714 correction and uses exactly the supervision the reviewer considers “light” (a handful of per-country  
715 scalars).

716 **(8) DiffPO-binary.** The spirit of DiffPO [Chen et al., 2025] adapted to binary decisions: we mix the  
717 vanilla probability with a country-conditioned target  $p_{\text{target}}(c, \text{cat})$  built directly from the country’s  
718 *public* human AMCE ( $p_{\text{aligned}} = (1 - \alpha) p_{\text{van}} + \alpha p_{\text{target}}$ ), with  $\alpha \in [0, 1]$  fit per country on the  
719 calibration split. Because this baseline *consumes* the evaluation target at inference, it is an upper  
720 bound on what any mixing of the public human AMCE with vanilla probabilities can achieve.

721 **Takeaway.** DISCA is compared against all three on the same twenty-country grid; because baselines  
722 (7) and (8) use *population-level* supervision while DISCA uses only *within-country persona dis-*  
723 *agreement* (never touching the human AMCE at inference), the comparison is deliberately favourable  
724 to the baselines. The per-country results are in Table 16: DISCA wins **18 of 20** countries against  
725 MC-Dropout, **19 of 20** against per-country temperature scaling, and **18 of 20** against margin scaling.  
726 Reference implementations of all three baselines are released with the code archive accompanying  
727 this submission.

Table 16: Phi-4 (14B) per-country head-to-head: DISCA vs. three inference-time baselines on the same 20-country test split. Per-country held-out MIS ( $\downarrow$ , lower is better). **Bold** cells mark the best of {DISCA, MC-Dropout,  $T_c$ ,  $m_c$ } per country. DiffPO-binary is omitted: as a population-target replay it scores  $\approx 0$  MIS by construction (each country’s prediction trivially copies the country’s human AMCE) and is not a comparable inference-time baseline.

| Country                 | DISCA (ours) | MC-Dropout   | $T_c$ scale | Margin $m_c$ |
|-------------------------|--------------|--------------|-------------|--------------|
| ARG                     | <b>0.389</b> | 0.412        | 0.471       | 0.453        |
| BGD                     | <b>0.368</b> | 0.379        | 0.535       | 0.535        |
| BRA                     | 0.299        | 0.306        | 0.354       | <b>0.298</b> |
| CHN                     | <b>0.214</b> | 0.367        | 0.528       | 0.528        |
| COL                     | 0.438        | <b>0.389</b> | 0.501       | 0.469        |
| DEU                     | <b>0.255</b> | 0.376        | 0.580       | 0.580        |
| ETH                     | <b>0.485</b> | 0.526        | 0.655       | 0.655        |
| GBR                     | <b>0.319</b> | 0.407        | 0.576       | 0.576        |
| IDN                     | <b>0.274</b> | 0.452        | 0.542       | 0.542        |
| IRN                     | 0.530        | 0.538        | 0.382       | <b>0.353</b> |
| JPN                     | <b>0.195</b> | 0.371        | 0.327       | 0.327        |
| KGZ                     | 0.393        | <b>0.361</b> | 0.538       | 0.538        |
| MEX                     | <b>0.420</b> | 0.445        | 0.509       | 0.506        |
| MMR                     | <b>0.357</b> | 0.382        | 0.523       | 0.523        |
| MYS                     | <b>0.328</b> | 0.363        | 0.497       | 0.497        |
| ROU                     | <b>0.385</b> | 0.398        | 0.564       | 0.564        |
| SRB                     | <b>0.385</b> | 0.410        | 0.563       | 0.563        |
| THA                     | <b>0.313</b> | 0.342        | 0.494       | 0.494        |
| USA                     | <b>0.243</b> | 0.369        | 0.564       | 0.564        |
| VNM                     | <b>0.336</b> | 0.473        | 0.561       | 0.561        |
| <b>Macro</b>            | <b>0.346</b> | 0.403        | 0.513       | 0.506        |
| <b>Wins (DISCA vs.)</b> | -            | 18/20        | 19/20       | 18/20        |

## 728 A10 Hyperparameters and Sensitivity Sweeps

729 This appendix collects the full hyperparameter specification of DISCA together with the sensitivity  
730 sweeps that justify the choices. We organise the material into four parts: a robustness summary that  
731 consolidates the headline checks, the canonical hyperparameter table with validation methodology,  
732 the per-axis temperature sweeps, and an extended sensitivity sweep over the four principal control  
733 knobs plus persona count and IS budget.

## A10.1 Robustness Summary

Table 17 consolidates the nine independent robustness checks referenced in §5. Each row states one assumption behind DISCA, the test that probes it, the threshold that would falsify the assumption, and the observed value. Per-axis sweeps and full per-country tables for each row appear in the subsections below and in Appendix A17.

Table 17: Robustness suite. Each test independently probes one assumption behind DISCA; the bootstrap confidence interval on JSD is  $\pm 0.004$ .

| Claim                                       | Test                                                         | Threshold    | Observed      |
|---------------------------------------------|--------------------------------------------------------------|--------------|---------------|
| Dataset preprocessing irrelevant            | JSD range over 6 configs                                     | $< 0.010$    | <b>0.0034</b> |
| $T_{\text{dec}}$ insensitivity              | JSD span (Sweep A)                                           | $< 0.010$    | <b>0.0074</b> |
| Uniform $T_{\text{cat}}$ insensitivity      | JSD span (Sweep B)                                           | $< 0.010$    | <b>0.0073</b> |
| Per-category $T_{\text{cat}}$ insensitivity | JSD span (Sweep C)                                           | $< 0.010$    | <b>0.0077</b> |
| PT-IS $>$ best deterministic shift          | $\Delta\text{JSD}(\text{CS-Clamp} \rightarrow \text{PT-IS})$ | $> 0$        | <b>0.003</b>  |
| WVS weighting alone insufficient            | $\text{ARGS-WVS} \succ \text{ARGS-Unif}$                     | $< 8/15$     | <b>3/15</b>   |
| PT-IS kernel $\equiv$ ARGS-PT               | ARGS-PT best/tied                                            | $\geq 10/15$ | <b>13/15</b>  |
| Cross-lingual pipeline generalises          | end-to-end runs                                              | 8/8          | <b>8/8</b>    |

## A10.2 Default Hyperparameters and Validation

Table 18: DISCA hyperparameters. Prospect Theory parameters follow Kahneman and Tversky [1979]; other defaults are validated on the synthetic pool described below.

| Parameter                                   | Symbol                  | Value                                                |
|---------------------------------------------|-------------------------|------------------------------------------------------|
| Persona agents                              | $N$                     | 4                                                    |
| IS samples / pass                           | $K_{\text{half}}$       | 64 (total $2K_{\text{half}} = 128$ )                 |
| Reliability scale                           | $s$                     | 0.04                                                 |
| Perturbation std. dev.                      | $\sigma$                | 0.3                                                  |
| Cooperation weight                          | $\lambda_{\text{coop}}$ | 0.7                                                  |
| IS softmax temp.                            | $\eta$                  | 0.5                                                  |
| ESS guard threshold                         | $\rho_{\text{eff}}$     | 0.1                                                  |
| PT curvature                                | $\alpha = \beta$        | 0.88                                                 |
| PT loss aversion                            | $\kappa$                | 2.25                                                 |
| Decision temp.                              | $T_{\text{dec}}$        | 0.5                                                  |
| $T_{\text{cat}}$ (Species / Gender / other) | -                       | 4.0 / 3.5 / 1.5                                      |
| Default logit divisor                       | $T_{\text{logit}}$      | 3.0                                                  |
| ESS anchor blend                            | -                       | on (off: $\bar{\delta} + \delta_{\text{IS}}^*$ only) |

**Implementation.** The dual-pass controller follows the equations in §3. Optional environment overrides can change  $K_{\text{half}}$ , reliability scale  $s$ , and whether the ESS-anchor blend is active; such flags must be applied before controller initialisation when running ablations.

Non-PT hyperparameters ( $\sigma$ ,  $\lambda_{\text{coop}}$ ,  $\eta$ ,  $T_{\text{dec}}$ ,  $T_{\text{cat}}$ ) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration.  $T_{\text{cat}}$  values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (Appendix A10, Appendix A17) to show conclusions are stable beyond one tuned point.

## A10.3 Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies  $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$ ; Sweep B replaces the per-category



temperatures by a single uniform  $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$ ; Sweep C varies the “Others” bucket  $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$  while holding  $T_{\text{cat}}[\text{Species}]=4.0$ ,  $T_{\text{cat}}[\text{Gender}]=3.5$ . All three sweeps yield JSD spans below 0.010 (0.0074, 0.0073, 0.0077 respectively), and in every sweep the DISCA default ( $\dagger$ ) is strictly *more conservative* than the empirically best value ( $\Delta\text{JSD} \in \{-0.0061, -0.0021, -0.0037\}$ ). This rules out any concern that results have been temperature-tuned to the benchmark.

Table 19: Temperature sensitivity. Defaults ( $\dagger$ ) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

| Sweep                       | Setting        | JSD $\downarrow$ | Pearson $r \uparrow$ | $\Delta\text{JSD}$ |
|-----------------------------|----------------|------------------|----------------------|--------------------|
| A: $T_{\text{dec}}$         | 0.10           | 0.0502           | 0.662                | +0.0013            |
|                             | 0.25           | 0.0491           | 0.658                | +0.0002            |
|                             | 0.50 $\dagger$ | 0.0489           | 0.630                | -                  |
|                             | 0.75           | 0.0476           | 0.636                | -0.0013            |
|                             | 1.00           | 0.0466           | 0.641                | -0.0023            |
|                             | 2.00 (best)    | 0.0428           | 0.627                | -0.0061            |
|                             | JSD span       | <b>0.0074</b>    |                      |                    |
| B: Uniform $T_{\text{cat}}$ | 0.5            | 0.0521           | 0.642                | +0.0052            |
|                             | 1.0            | 0.0492           | 0.673                | +0.0023            |
|                             | 1.5 $\dagger$  | 0.0469           | 0.700                | -                  |
|                             | 2.0            | 0.0461           | 0.707                | -0.0008            |
|                             | 3.0            | 0.0454           | 0.707                | -0.0015            |
|                             | 4.0            | 0.0451           | 0.701                | -0.0018            |
|                             | 6.0 (best)     | 0.0448           | 0.684                | -0.0021            |
|                             | JSD span       | <b>0.0073</b>    |                      |                    |
| C: Others $T_{\text{cat}}$  | 0.75           | 0.0526           | 0.566                | +0.0039            |
|                             | 1.00           | 0.0512           | 0.589                | +0.0025            |
|                             | 1.50 $\dagger$ | 0.0487           | 0.634                | -                  |
|                             | 2.00           | 0.0475           | 0.660                | -0.0012            |
|                             | 3.00           | 0.0458           | 0.693                | -0.0029            |
|                             | 4.00 (best)    | 0.0450           | 0.706                | -0.0037            |
|                             | JSD span       | <b>0.0077</b>    |                      |                    |

The monotonic JSD reduction with larger  $T_{\text{dec}}$  reflects the role of  $T_{\text{dec}}$  in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However,  $r$  peaks near  $T_{\text{dec}}=1.0$  and declines at 2.0, indicating a trade-off between distributional alignment (JSD) and rank-order alignment ( $r$ ); we keep  $T_{\text{dec}}=0.5$  as a deliberately conservative operating point.

#### A10.4 Extended Hyperparameter Sensitivity

We probe the sensitivity of DISCA to the four principal hyperparameters:  $s$  (reliability gate scale),  $\lambda_{\text{coop}}$  (individual-vs-consensus weight),  $\sigma$  (IS proposal floor), and  $T_{\text{cat}}$  (uniform logit-temperature scaling). For each axis we sweep five values on a three-country panel (USA, VNM, DEU) and hold the other three axes at their defaults; the  $s$  axis directly probes how the choice of reliability scale in Eq. 6 affects downstream MIS.

Table 20: Hyperparameter sensitivity ranges on the three-country panel (Phi-4,  $n=250/\text{country}$ ). Reported: min/max MIS across the five grid points relative to the default; MIS is stable within a narrow 0.0037–0.0090 span on every axis, so the gains in Table 3 are not the product of a particular hyperparameter choice.

| Axis                    | Default     | Grid                                                                        | Min MIS | Max MIS | $\Delta$ |
|-------------------------|-------------|-----------------------------------------------------------------------------|---------|---------|----------|
| $s$                     | 0.04        | {0.01, 0.02, 0.04, 0.08, 0.16}                                              | 0.4091  | 0.4164  | 0.0073   |
| $\lambda_{\text{coop}}$ | 0.70        | {0.30, 0.50, 0.70, 0.85, 0.95}                                              | 0.4085  | 0.4173  | 0.0088   |
| $\sigma$                | 0.30        | {0.10, 0.20, 0.30, 0.45, 0.60}                                              | 0.4084  | 0.4122  | 0.0037   |
| $T_{\text{cat}}$ scale  | $1.0\times$ | { $0.33\times$ , $0.67\times$ , $1.0\times$ , $1.33\times$ , $1.67\times$ } | 0.4060  | 0.4149  | 0.0090   |

772 **Why these four axes.**  $s$  controls the steepness of the reliability gate (Eq. 6);  $\lambda_{\text{coop}}$  trades off  
773 individual-persona PT utility against the consensus utility;  $\sigma$  floors the IS proposal scale so the  $N=4$   
774 empirical std cannot collapse to zero; and  $T_{\text{cat}}$  scales the whole family of per-category decision  
775 temperatures (preserving relative balance)-a global sharpening / flattening knob on the decision logits.  
776 Per-category temperatures are stress-tested separately in the temperature sensitivity sweep above.

777 **Persona count, IS budget, and per-category vs. global  $T_{\text{cat}}$ .** We additionally swept three op-  
778 erational knobs on the same USA/VNM/DEU panel. **Persona count** ( $N \in \{2, 3, 4, 5, 6\}$ , Ta-  
779 ble 21):  $N=4$  is optimal at macro MIS 0.4051, with degradation at both smaller and larger pan-  
780 els (0.4252 at  $N=2$ , 0.4179 at  $N=6$ ), supporting the four-persona design. **IS budget** ( $K_{\text{half}} \in$   
781 {8, 16, 32, 64, 128, 192}, Table 22): performance is flat across an order of magnitude (best 0.4092 at  
782  $K=16$ ; 0.4103 at  $K=64$ ), indicating the default  $K_{\text{half}}=64$  is in the stable region rather than over-  
783 tuned. **Global vs. per-category  $T_{\text{cat}}$**  (Table 23): the per-category default outperforms *every* global  
784 schedule we tested (0.4098 vs. all seven global values in {1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0}, which land  
785 in 0.4119–0.4138). The advantage is small in absolute terms but consistent.

Table 21: Persona-count sweep on the USA/VNM/DEU panel (Phi-4,  $n=250/\text{country}$ ). The four-persona default minimises macro MIS; larger panels add latency without alignment benefit.

| $N$      | Macro MIS $\downarrow$ | Std (across countries) | Flip rate | sec/scenario |
|----------|------------------------|------------------------|-----------|--------------|
| 2        | 0.4252                 | 0.1045                 | 0.275     | 0.089        |
| 3        | 0.4194                 | 0.0945                 | 0.273     | 0.100        |
| <b>4</b> | <b>0.4051</b>          | 0.0829                 | 0.276     | 0.111        |
| 5        | 0.4107                 | 0.0745                 | 0.276     | 0.120        |
| 6        | 0.4179                 | 0.0919                 | 0.245     | 0.130        |

Table 22: IS-budget sweep ( $K_{\text{half}}$  samples per pass; total per scenario is  $K_{\text{total}}=2K_{\text{half}}$ ). Macro MIS is flat across an order of magnitude. The mean reliability weight  $\bar{r}$  (gate column) grows monotonically with  $K$ , confirming the gate becomes more confident as the IS proposal cloud densifies.

| $K_{\text{half}}$ | $K_{\text{total}}$ | Macro MIS $\downarrow$ | Flip rate | $\bar{r}$ (gate) | sec/scen |
|-------------------|--------------------|------------------------|-----------|------------------|----------|
| 8                 | 16                 | 0.4114                 | 0.284     | 0.684            | 0.121    |
| 16                | 32                 | 0.4092                 | 0.271     | 0.736            | 0.114    |
| 32                | 64                 | 0.4104                 | 0.278     | 0.766            | 0.114    |
| 64                | 128                | 0.4103                 | 0.280     | 0.793            | 0.113    |
| 128               | 256                | 0.4147                 | 0.274     | 0.814            | 0.116    |
| 192               | 384                | 0.4126                 | 0.273     | 0.827            | 0.112    |

Table 23: Global vs. per-category  $T_{\text{cat}}$  on the USA/VNM/DEU panel. The per-category default (decision 4.0, social 3.5, non-social 1.5) outperforms every global value we tested. The gap is small ( $\leq 0.004$ ) but uniformly directional, supporting category-specific decision temperatures over a single shared scalar.

| $T_{\text{cat}}$ schedule                     | Macro MIS ↓   | Std    | Flip rate |
|-----------------------------------------------|---------------|--------|-----------|
| <b>per-category default (4.0 / 3.5 / 1.5)</b> | <b>0.4098</b> | 0.0881 | 0.278     |
| global $T_{\text{cat}} = 1.00$                | 0.4120        | 0.0921 | 0.292     |
| global $T_{\text{cat}} = 1.50$                | 0.4119        | 0.0838 | 0.304     |
| global $T_{\text{cat}} = 2.00$                | 0.4138        | 0.0778 | 0.301     |
| global $T_{\text{cat}} = 2.50$                | 0.4134        | 0.0774 | 0.296     |
| global $T_{\text{cat}} = 3.00$                | 0.4127        | 0.0749 | 0.303     |
| global $T_{\text{cat}} = 3.50$                | 0.4138        | 0.0689 | 0.306     |
| global $T_{\text{cat}} = 4.00$                | 0.4128        | 0.0734 | 0.302     |

## A11 Scenario-Slice Release and Category-Cap Sensitivity

Section 4.1 describes our preprocessing pipeline, which includes a per-category cap at 80 scenarios to prevent any single moral dimension from dominating the evaluation pool. To give full reproducibility and to confirm that the cap is not an artefact-generating shortcut, we provide two additional artifacts:

- **Exact scenario-slice release.** The deterministic seed-42 scenario IDs for each of the twenty countries are released alongside the code archive, allowing any reader to reproduce the evaluation slice bit-for-bit without rerunning the preprocessing pipeline.
- **No-cap sensitivity run.** We re-run Phi-4 DISCA on the twenty-country grid with the per-category cap disabled. This run executes 440 scenarios per country (vs. the capped slice) and yields macro MIS 0.4051, macro JSD 0.0662, and mean Pearson  $r$  0.2878; performance remains in the same operating regime, indicating the cap is primarily a variance-control device rather than a load-bearing preprocessing choice.

Both runs use the same seed, native-language framing, and hyperparameters as the main Phi-4 experiment.

## A12 Theoretical Grounding

DISCA sits at the intersection of three theoretical traditions, each supplying one ingredient that the others lack. *Control as probabilistic inference* [Levine, 2018, Todorov, 2009] recasts optimal decision-making as posterior inference over an intention distribution; path-integral steering [Williams et al., 2018] instantiates that view with softmax-weighted perturbations that provably approximate the KL-regularised optimal control under mild regularity conditions [Todorov, 2009]. This is what licences us to read our Eq. 5 as an inference step rather than an arbitrary scalar interpolation, and it motivates the Gaussian proposal around  $\bar{\delta}$  as the entropy-maximal default in the absence of stronger priors. *Prospect Theory* [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] then supplies the utility shape: the key empirical regularity—that human decision makers weight a loss of magnitude  $x$  roughly  $\kappa \approx 2.25$  times more than a gain of the same magnitude—is exactly the asymmetry we want an LLM cultural controller to respect, because a correction that silences one demographic persona is worse than a correction that fails to help it. Recent evidence that prospect-theoretic biases emerge naturally in LLM decision-making under risk [Payne, 2025] strengthens the match: the asymmetric gain/loss structure we impose *mirrors* one the models themselves already exhibit, rather than introducing an alien objective. *Importance-sampling (IS) theory* [Elvira and Martino, 2021] finally supplies the variance-control machinery, and it is here that self-normalised IS is known to be biased at finite  $K$ ; the dual-pass construction we adopt is a light-weight bootstrap-style variance probe [Efron, 1979] that uses the same compute budget to detect instability rather than silently accept it, paired with an effective-sample-size guard grounded in design-effect theory [Kish, 1965]. We treat the dual-pass gate as a variance heuristic rather than a formal unbiasedness certificate; formal finite- $K$  bounds would require stronger assumptions on the logit landscape than we are willing to claim (Appendix A14).

## A13 Design Rationale: Why Each Component

This appendix collects the full justification for the three design choices previewed in §1 and §3: the four-persona WVS panel, the Prospect-Theory IS kernel, and the dual-pass reliability gate.

**Why a panel, and why the World Values Survey?** A single “prompt-as-country” encoder-e.g., “*Answer as a Japanese person*”—collapses a distribution to a point estimate and discards the very spread the correction stage needs. Cultural-value research has long argued against this framing: both Inglehart-Welzel’s modernisation theory [Inglehart and Welzel, 2005] and Schwartz’s refined theory of basic individual values [Schwartz et al., 2012] find within-country variance on any single value dimension comparable to between-country variance, with demographic cleavages—most prominently age cohort—driving the largest slice. The Moral Machine analysis confirms this specifically for trolley preferences: generational differences in how respondents weight youth, social status, and species cut across every country cluster [Awad et al., 2018, Kleiman-Weiner et al., 2017]. WVS Wave 7 [Haerpfer et al., 2022] is the largest public instrument that reports *individual-level* microdata with the relevant age, gender, and value coverage, making it the natural substrate for data-driven personas rather than author-crafted stereotypes—a well-documented failure mode of prompt-based persona work [Deshpande et al., 2023].

**Why  $N=4$  specifically?** Four is the smallest panel that simultaneously covers the three demographic axes documented to carry most within-country variance—young ( $<36$ ), middle-aged ( $36\text{--}55$ ), older ( $>55$ )—plus a country-wide aggregate that anchors the ensemble against cohort-specific sampling noise. Three personas would sacrifice the population anchor and leave the consensus  $\bar{\delta}$  vulnerable to a single outlier cohort; five or more would duplicate coverage of the same axis without adding a new source of variance (WVS does not resolve finer cohorts reliably for every country). The cohort structure matches the statistical-survey tradition of stratifying by age before any country-level estimate [Kish, 1965]. Empirically, the ablation in §6 confirms the ensemble is load-bearing: removing it costs  $-0.183$  Pearson  $r$  and  $+0.083$  MIS.

**Why Prospect Theory specifically, not a linear or quadratic aggregator?** Three independent considerations pick out the Kahneman-Tversky value function [Kahneman and Tversky, 1979, Tversky and Kahneman, 1992] over simpler alternatives. (1) *Asymmetric harm*. In a cultural panel, a correction that improves three personas while making one worse is not equivalent to one that helps all four equally: the former silences a demographic group the survey says exists. Linear averaging is oblivious to this asymmetry; quadratic aggregation treats  $+x$  and  $-x$  as equally valuable and costly. Loss aversion with  $\kappa \approx 2.25$  is the textbook object for encoding “harm is worse than equal-sized help”. (2) *Diminishing sensitivity matches the scale of the cultural signal*. Gain curvature  $\alpha = 0.88 < 1$  compresses large persona gains relative to small ones, preventing one outlier persona (perhaps from sparse WVS coverage) from dominating the utility [Tversky and Kahneman, 1992]. (3) *Empirical compatibility with the base model*. Recent work shows that LLMs exhibit prospect-theoretic biases in decision-making under risk, with loss-aversion coefficients in the same neighbourhood as human respondents [Payne, 2025]. While we do not claim that logit-gap changes in an LLM are psychologically equivalent to monetary gains and losses in human subjects—the PT value function is used here as a *controller aggregation kernel*, not as a cognitive model of the LLM—the empirical compatibility means we impose a utility shape that does not fight the model’s own implicit asymmetries. Auxiliary runs with linear reward kernels and WVS-weighted linear kernels on the same pipeline confirm that the PT shape outperforms symmetric alternatives, and the robustness sweeps in Appendix A10 show that the method’s conclusions are stable across three logit-shift kernels, so the specific  $\kappa = 2.25$  is a well-motivated default rather than a sensitive tuning knob.

**Why importance sampling at all?** The cultural correction we seek is the argmax of a utility that values each persona’s alignment and penalises corrections that make any of them worse [Levine, 2018]. Closed-form optimisation is unavailable because the utility is non-convex (PT’s kink at zero, the  $\ell_1$ -style gain definition in Eq. 13), so we approximate the optimal control distribution with a path-integral-style softmax over perturbations [Williams et al., 2018, Todorov, 2009]. The Gaussian proposal  $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$  plays the role of an entropy-maximal default, and the softmax weights  $w_k \propto \exp(U_{\text{total}}/\eta)$  implement the KL-regularised Boltzmann posterior balancing utility against

deviation from the consensus. This control-as-inference reading licenses Eq. 5 as an approximate Bayesian update rather than arbitrary scalar interpolation.

**Why dual-pass, rather than more samples or a hard threshold?** Self-normalised IS is a biased estimator at finite sample size [Elvira and Martino, 2021]: the bias shrinks as  $O(1/K)$  but does not reveal itself to the caller, so “more samples” hides instability rather than flagging it. Two independent replicates of size  $K_{\text{half}}$  using the same total budget  $K=2K_{\text{half}}$  trade a small increase in per-pass variance for a *direct, scenario-level probe* of that instability: the squared gap  $(\delta_1^* - \delta_2^*)^2$  is a one-dimensional bootstrap-style variance estimator [Efron, 1979] whose small values license the correction and whose large values veto it automatically, without a hand-tuned threshold. The exponential map  $r = \exp(-(\delta_1^* - \delta_2^*)^2/s)$  is preferred over a hard indicator because it preserves a single differentiable code path and its decay matches the shape of a Gaussian likelihood ratio under independent noise. An effective-sample-size guard [Kish, 1965] on each pass ( $N_{\text{eff}}/K_{\text{half}} < \rho_{\text{eff}}$ ) zeros the offset entirely when the weights collapse, providing a second, coarser safety net below the smooth reliability gate. We emphasise that this construction is a variance heuristic, not a formal unbiasedness certificate; finite- $K$  bounds would require stronger assumptions on the logit landscape than we are willing to claim (Appendix A14).

## A14 PT-IS, Sampling, and Dual-Pass Reliability

### A14.1 Gain definitions

For candidate  $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$ , the per-persona and consensus gains are:

$$g_{i,k} = |\delta_{\text{base}} - \delta_i| - |\tilde{\delta}_k - \delta_i|, \quad (13)$$

$$g_{\text{cons},k} = |\delta_{\text{base}} - \bar{\delta}| - |\tilde{\delta}_k - \bar{\delta}|. \quad (14)$$

$g_{i,k} > 0$  means the candidate is closer to persona  $i$  than the base model;  $g_{\text{cons},k}$  is the analogous quantity for the persona consensus. These enter Eq. 5 after  $\sigma$ -normalisation. The IS update for each pass is  $\delta_m^* = \sum_k w_k^{(m)} \epsilon_k$  with  $w_k^{(m)} = \text{softmax}(U_{\text{total}}(\epsilon_k)/\eta)$  and temperature  $\eta = 0.5$ . An ESS guard zeros  $\delta_m^*$  when the Kish effective sample size  $N_{\text{eff}}/K_{\text{half}}$  falls below  $\rho_{\text{eff}} = 0.1$ .

### A14.2 Utility shape and the omitted explicit KL-on- $\epsilon$ term

The scalar decision gap  $\delta$  acts as a one-dimensional state and the perturbations  $\epsilon_k$  as control inputs at a single horizon step. Under the path-integral / free-energy view [Williams et al., 2018], softmax weights  $w_k \propto \exp(U_{\text{total}}/\eta)$  can be read as targeting a Boltzmann distribution over candidate perturbations. **Our released code does not add** a separate quadratic penalty  $\propto \epsilon_k^2/(2\sigma^2)$  to  $U_{\text{total}}$  in Eq. 5: the released controller matches it exactly, and the Gaussian proposal  $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$  already limits how far  $\tilde{\delta}_k$  can move from  $\bar{\delta}$ . An optional explicit KL-on-control term remains a straightforward extension if future work needs sharper tail control on  $|\epsilon_k|$ .

### A14.3 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. PT-weighted IS performs an *adaptive, scenario-specific* correction via nonlinear importance weighting. Auxiliary runs with deterministic consensus-only shifts and WVS-linear reward kernels underperform the Prospect-Theory kernel on the same pipeline; we omit utility-shape rows from Table 5 for clarity.

### A14.4 Why split $K$ into two passes?

Self-normalised IS is biased at finite sample size; two independent halves with the *same* total budget trade per-pass variance for a direct probe of sampling instability. The reliability map  $r = \exp(-(\delta_1^* - \delta_2^*)^2/s)$  is intentionally smooth-unlike a hard veto it preserves a single code path and down-weights only when the two estimates contradict each other strongly. It should be read as a variance heuristic, not as a statistical guarantee of unbiasedness; formal finite- $K$  bounds would require stronger assumptions on the logit landscape than we are willing to claim here.

## A14.5 Positional Debiasing: Implementation Detail

The two-pass debiasing of §3.4 is implemented as a text-level rewrite of the scenario *after* native-language framing. To avoid the trivial bug of double-substitution (replacing LEFT → RIGHT immediately followed by RIGHT → LEFT would leave every label as LEFT), we use a placeholder-based three-step swap, applied independently for the lane-label pair and the group-label pair:

1. Replace every occurrence of the left label with a unique placeholder token (a non-printing sentinel).
2. Replace every occurrence of the right label with the left label.
3. Replace the placeholder with the right label.

The same procedure is applied to Group A ↔ Group B in the native script. Logit gaps ( $\delta_{\text{base}}, \delta_i$ ) from the two orders are combined *before* any nonlinear step,  $\delta \leftarrow (\delta^{(\text{orig})} - \delta^{(\text{swap})})/2$ , matching the released controller (debias first, then PT-IS and dual-pass merge). This guarantees cancellation of additive positional bias in logit space; the final  $p_{\text{spare}}$  applies  $\sigma(\cdot)$  only after that correction.

## A15 Persona Construction Details

This appendix specifies how the four cultural personas per country are constructed and provides empirical support that the construction is grounded in human survey data rather than authored ad hoc. §A15.1 reproduces two personas verbatim from `generate_wvs_persona()`; §A15.2 documents the WVS Wave 7 processing pipeline; §A15.3 links the ten WVS cultural dimensions to the six MultiTP moral attributes both thematically and empirically; and §A15.4 shows that DISCA is robust to the choice of the fourth (anchor) persona.

### A15.1 Example Personas

Each country yields four personas: three age-cohort personas (*young, middle, older*) generated from the country-and-cohort-specific WVS-7 means, plus a population-wide aggregate that serves as a demographic anchor. We illustrate the output with one English persona (United States, young-adult cohort) and one Vietnamese persona (Vietnam, older-adult cohort). The Vietnamese block is reproduced verbatim from `generate_wvs_persona()` and is followed by a sentence-by-sentence English gloss for non-Vietnamese readers.

#### United States, young-adult cohort (English, verbatim).

*You are a young adult from the United States, in your 20s and early 30s. Your worldview is shaped by the cultural values prevalent in your community. On matters of faith you are moderately religious. On raising children you are firmly oriented toward independence and imagination. On contested moral choices you are morally conservative on contested issues. In your dealings with strangers you have a guarded attitude toward strangers. Civically you are an active political participant who signs petitions, joins boycotts and takes part in lawful demonstrations. You are moderately proud of your country. Overall you are rather happy with your life. On the role of women in society you are moderately egalitarian on gender roles. In what you prioritise in life you are leaning materialist, prioritising economic and physical security. Toward people unlike yourself you are highly tolerant of outgroups such as immigrants, minorities and people with different lifestyles. When you face a moral dilemma, you weigh the choices through this set of values and answer in a way that is consistent with the worldview above.*

#### Vietnam, older-adult cohort (Vietnamese, verbatim).

*Bạn là một người cao tuổi đến từ Việt Nam, trên 60 tuổi. Thế giới quan của bạn được định hình bởi các giá trị văn hóa phổ biến trong cộng đồng của bạn. Về vấn đề đức tin, bạn có niềm tin tôn giáo ở mức trung bình. Trong việc nuôi dạy con cái, bạn nghiêng về sự vâng lời và đức tin tôn giáo. Về các lựa chọn đạo đức gây tranh cãi, bạn bảo thủ về mặt đạo đức trong các vấn đề gây tranh cãi. Trong giao tiếp với người lạ, bạn có thái độ dè dặt với người lạ. Về mặt công dân, bạn là một người thành thạo tham gia chính trị. Bạn rất tự hào về đất nước của bạn. Nhìn chung, bạn rất hài lòng với cuộc sống của bạn. Về vai trò của phụ*

970 nữ trong xã hội, bạn khá truyền thống về vai trò giới. Về những gì bạn ưu tiên trong cuộc  
 971 sống, bạn có xu hướng hậu vật chất. Đối với những người khác bạn, bạn rất thiếu khoan  
 972 dung với các nhóm bên ngoài. Khi bạn đối mặt với một tình huống khó xử về đạo đức, bạn  
 973 cân nhắc các lựa chọn thông qua tập hợp giá trị này và trả lời theo cách phù hợp với thể  
 974 giới quan đã nêu ở trên.

#### 975 Sentence-by-sentence English gloss.

976 *You are a senior citizen from Vietnam, over 60 years old. Your worldview is shaped by*  
 977 *the cultural values prevalent in your community. On matters of faith you are moderately*  
 978 *religious. On raising children you are leaning toward obedience and religious faith. On*  
 979 *contested moral choices you are morally conservative on contested issues. In your dealings*  
 980 *with strangers you have a guarded attitude toward strangers. Civically you are an occasional*  
 981 *political participant. You are intensely proud of your country. Overall you are very happy*  
 982 *with your life. On the role of women in society you are somewhat traditional on gender*  
 983 *roles. In what you prioritise in life you are leaning post-materialist. Toward people unlike*  
 984 *yourself you are strongly intolerant of outgroups. When you face a moral dilemma, you*  
 985 *weigh the choices through this set of values and answer in a way that is consistent with the*  
 986 *worldview above.*

987 **Aggregate (fourth) persona.** The fourth agent is constructed from the country’s population-  
 988 wide WVS profile, pooling respondents across age cohorts. It serves as a demographic anchor:  
 989 by suppressing cohort-specific noise it ensures that the ensemble’s consensus target  $\delta$  reflects the  
 990 broadest available empirical signal rather than a single generational viewpoint. §A15.4 contrasts this  
 991 choice with a country-invariant utilitarian anchor and shows that DISCA’s macro-MIS is insensitive  
 992 to the substitution.

993 **Faithfulness disclaimer.** The descriptors above are read directly from the country’s WVS-7 means  
 994 through deterministic quartile cuts (Table 24); they are not hand-selected to match cultural stereotypes.  
 995 As a consequence, individual cohorts can present empirically grounded but locally counter-intuitive  
 996 combinations (for example, a U.S. young-adult that is *leaning materialist* and *morally conservative*  
 997 *on contested issues*), reflecting actual WVS-7 distributions rather than authorial bias.

#### 998 A15.2 WVS Data Processing Pipeline

999 We process WVS Wave 7 individual-level microdata following the ten-variable cultural-value scheme  
 1000 of Greco et al. [2026]. The pipeline is fully deterministic: given the inverted WVS-7 CSV and the  
 1001 country code, it returns the four personas used at inference time without any tunable hyperparameters.

- 1002 1. **Variable extraction:** For each respondent, we extract 10 value dimensions from WVS items  
 1003 (Table 24). The “inverted” WVS-7 CSV (suffix P) pre-flips Likert items so higher values  
 1004 consistently indicate the positive pole. Items without the P suffix (Q152, Q153, Q177–Q182)  
 1005 use original WVS coding.
- 1006 2. **Age cohort assignment:** Using birth year (Q261) and survey year (A\_YEAR), we compute  
 1007 age and assign to: *young* (<36), *middle* (36–55), *older* (>55). Respondents with birth year  
 1008 before 1900 or after 2010, or survey year before 2015, are excluded. Negative WVS codes  
 1009 (−1, −2, −4, −5: refusal / don’t know) are dropped; zero is retained for binary 0/1 items.
- 1010 3. **Cohort aggregation:** For each country and age cohort, each dimension’s score is the  
 1011 unweighted mean of all valid (respondent, item) values associated with that dimension —  
 1012 i.e., for multi-item dimensions (e.g., Q177–Q182 for moral acceptability) every respondent  
 1013 contributes one value per item to a single pooled mean. For single-item dimensions this  
 1014 reduces to the standard across-respondent mean.
- 1015 4. **Normalisation and descriptor mapping:** Raw dimension means are normalised to [0, 1]  
 1016 via  $(\text{value} - \text{lo}) / (\text{hi} - \text{lo})$  where (lo, hi) is the WVS scale range, then directionally flipped  
 1017 so higher = positive pole. Four-level descriptors are assigned by quartile cuts:  $\geq 0.75$   
 1018 (strong positive),  $\geq 0.50$ ,  $\geq 0.25$ ,  $< 0.25$  (strong negative). See Table 24.
- 1019 5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use  
 1020 manually authored native-language personas grounded in area-studies literature, covering  
 1021 the same demographic structure (3 age cohorts + 1 population-wide aggregate).

Table 24: The 10 WVS cultural-value dimensions used for persona generation, following Greco et al. [2026]. All dimensions use uniform quartile cuts on the  $[0, 1]$ -normalised score:  $\geq 0.75$  (descriptor 1, strongest positive pole),  $\geq 0.50$  (2),  $\geq 0.25$  (3),  $< 0.25$  (4, strongest negative pole). The “direction” column indicates whether higher raw WVS values align with (+) or oppose (−) the positive pole.

| Dimension        | WVS items              | Range | Dir. | Descriptors ( $\geq 0.75$ / $\geq 0.50$ / $\geq 0.25$ / $< 0.25$ ) |
|------------------|------------------------|-------|------|--------------------------------------------------------------------|
| Religiosity      | Q6P                    | 1–4   | +    | deeply / moderately religious / somewhat / highly secular          |
| Child-rearing    | Q17P                   | 0–1   | −    | independence-imagination $\leftrightarrow$ obedience-faith         |
| Moral accept.    | Q177–Q182              | 1–10  | +    | very permissive $\leftrightarrow$ strictly opposed                 |
| Social trust     | Q57P                   | 1–2   | +    | very high trust $\leftrightarrow$ deep distrust                    |
| Polit. particip. | Q199P, Q200P           | 1–3   | +    | active participant $\leftrightarrow$ non-participant               |
| National pride   | Q254P                  | 1–4   | +    | very proud $\leftrightarrow$ not proud                             |
| Happiness        | Q46P                   | 1–4   | +    | very happy $\leftrightarrow$ not happy                             |
| Gender equality  | Q29P, Q30P, Q31P, Q33P | 1–4   | −    | strongly egalitarian $\leftrightarrow$ traditional roles           |
| Materialism      | Q152, Q153             | 1–3   | +    | post-materialist $\leftrightarrow$ materialist                     |
| Tolerance        | Q19P–Q23P              | 0–1   | −    | very tolerant $\leftrightarrow$ intolerant                         |

### 1022 A15.3 WVS-to-Trolley Dimension Linkage

1023 The linkage between the ten WVS cultural dimensions and the six MultiTP moral dimensions operates  
1024 through two complementary mechanisms.

1025 **Direct thematic overlap.** WVS *gender equality* maps onto the Gender dimension—societies  
1026 scoring higher exhibit weaker female-sparing preferences [Awad et al., 2018]; *religiosity* correlates  
1027 with Species through religious anthropocentrism and human exceptionalism; and *moral acceptability*  
1028 relates to tolerance for trade-offs along the Fitness and Social Value dimensions.

1029 **Indirect modulation.** Dimensions such as *social trust*, *child-rearing values*, and *materialism*  
1030 *orientation* reshape the moral *frame* the persona adopts: post-materialist personas favour utilitarian  
1031 calculi, while obedience-oriented child-rearing values up-weight hierarchical social roles (affecting  
1032 Age and Social Value). Importantly, DISCA does not require an exact causal mapping between WVS  
1033 and MultiTP dimensions; the ten-dimensional persona ensemble produces diversity in logit-space  
1034 gaps, and the importance-sampling stage selects the correction that balances collective utility.

1035 **Empirical validation of the WVS→AMCE pathway.** Table 25 regresses each of the six MultiTP  
1036 human AMCE dimensions on the 10 WVS cultural features across the 20-country panel (standardised  
1037 OLS). The 10-feature WVS profile explains  $R^2 \in [0.55, 0.69]$  of human AMCE variance, with  
1038  $|\beta| > 0.3$  in 31 of 60 cells. This is direct evidence that WVS-grounded persona construction is not  
1039 arbitrary: the cultural axes our personas inherit correlate strongly with the moral preferences DISCA  
1040 must reproduce.

Table 25: WVS features as predictors of human AMCE dimensions. OLS with standardised predictors. Coefficients are standardised ( $\beta$ ); bold =  $|\beta| > 0.3$ .  $R^2_{\text{adj}}$  measures how well the 10 WVS cultural dimensions explain each moral preference dimension across the 20-country panel.

| AMCE dim       | relig        | child        | moral        | socia        | polit        | natio        | happi        | gende        | mater        | toler        | $R^2$ | $R^2_{\text{adj}}$ |
|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------|--------------------|
| Species        | <b>+0.47</b> | +0.12        | <b>-0.65</b> | <b>+0.60</b> | +0.08        | +0.16        | -0.02        | <b>-0.74</b> | -0.30        | <b>+0.40</b> | 0.57  | 0.09               |
| Gender         | <b>+0.69</b> | <b>+0.37</b> | -0.18        | <b>+0.41</b> | <b>-0.44</b> | <b>-0.39</b> | <b>+0.71</b> | <b>-0.81</b> | -0.20        | +0.06        | 0.66  | 0.29               |
| Age            | <b>-0.80</b> | -0.09        | <b>+0.61</b> | <b>-0.74</b> | +0.17        | +0.26        | -0.27        | <b>+0.62</b> | <b>+0.49</b> | <b>-0.37</b> | 0.55  | 0.05               |
| Fitness        | +0.28        | -0.23        | -0.16        | -0.12        | -0.17        | <b>+0.43</b> | +0.09        | <b>-1.46</b> | -0.21        | <b>+0.53</b> | 0.60  | 0.15               |
| Social Value   | -0.27        | -0.30        | <b>+0.40</b> | <b>-0.40</b> | <b>-0.37</b> | <b>+0.58</b> | +0.17        | <b>-0.31</b> | -0.12        | -0.15        | 0.69  | 0.35               |
| Utilitarianism | <b>+1.11</b> | <b>+0.39</b> | +0.07        | <b>+0.59</b> | -0.21        | -0.01        | +0.13        | <b>-0.76</b> | <b>+0.42</b> | +0.25        | 0.66  | 0.29               |

1041 Table 26 closes the loop with a causal leave-one-WVS-dim-out probe: for each WVS dimension  
1042  $d$ , we re-run DISCA with  $d$  removed from every persona and measure the change in per-MultiTP-  
1043 dim absolute error (positive cell = error rises when the dimension is removed = that dimension is  
1044 load-bearing for the corresponding moral attribute). The three largest positive couplings are *moral*  
1045 *acceptability*  $\rightarrow$  Social Value (+3.97 pp), *social trust*  $\rightarrow$  Social Value (+2.81 pp), and *child-rearing*  
1046 *values*  $\rightarrow$  Utilitarianism (+2.41 pp). Cells with the top-3 largest positive couplings per row are  
1047 bolded; rows with fewer than three positive cells are bolded only on those positive cells. Negative  
1048 cells are also informative: *tolerance of diversity*  $\rightarrow$  Social Value (−7.76) and *gender equality*  $\rightarrow$



Gender ( $-3.42$ ) indicate that, for those moral attributes, the corresponding WVS descriptor biases the persona ensemble *away* from the human target, so removing the descriptor improves accuracy. Crucially, DISCA does not require manual feature selection to handle these noisy dimensions: the loss-averse importance-sampling stage (Eq. 5) automatically down-weights candidate perturbations that worsen any persona’s alignment, effectively acting as a built-in noise filter. When a WVS dimension introduces bias for a particular moral attribute, the PT-IS utility penalises the resulting correction asymmetrically ( $\kappa=2.25\times$  for losses vs. gains), ensuring that noisy persona signals are suppressed rather than propagated. This self-correcting property is why the method remains robust despite imperfect WVS-to-trolley mappings—a design-level answer to the concern that the linkage is “indirect modulation” rather than causal.

Table 26: WVS-dim  $\times$  MultiTP-dim causal impact matrix. Cells are the mean increase in per-dim AMCE error (pp) when the WVS dimension is dropped from every persona, macro-averaged across the country panel. **Bold** cells mark the top-3 largest positive couplings per row (load-bearing WVS  $\rightarrow$  MultiTP links).

| WVS dim                 | Species_Humans | Gender_Female | Age_Young    | Fitness_Fit  | SocialValue_High | Utilitarianism_More |
|-------------------------|----------------|---------------|--------------|--------------|------------------|---------------------|
| religiosity             | <b>+0.66</b>   | <b>+1.65</b>  | -1.66        | +0.58        | <b>+1.02</b>     | -0.37               |
| child rearing           | -1.82          | <b>+1.29</b>  | +0.86        | <b>+1.30</b> | -0.33            | <b>+2.41</b>        |
| moral acceptability     | <b>+1.13</b>   | -0.02         | -1.56        | -0.34        | <b>+3.97</b>     | -1.69               |
| social trust            | +0.11          | <b>+1.43</b>  | -0.85        | <b>+0.51</b> | <b>+2.81</b>     | -1.88               |
| political participation | -1.24          | -0.82         | -0.63        | -0.36        | -0.16            | <b>+0.86</b>        |
| national pride          | -0.37          | -0.10         | -0.28        | <b>+0.52</b> | <b>+1.19</b>     | <b>+1.21</b>        |
| happiness               | -0.23          | -0.88         | <b>+0.13</b> | -1.69        | -0.11            | <b>+1.72</b>        |
| gender equality         | <b>+0.41</b>   | -3.42         | -2.90        | -1.11        | -1.87            | <b>+0.62</b>        |
| materialism orientation | <b>+1.35</b>   | -1.26         | <b>+0.90</b> | +0.04        | -0.36            | <b>+2.15</b>        |
| tolerance diversity     | -0.76          | -3.52         | <b>+1.95</b> | -0.59        | -7.76            | <b>+1.54</b>        |

#### A15.4 Sensitivity to the Fourth Persona

**Setup.** §3.2 pairs three age-cohort personas with a population-wide *aggregate* WVS profile as the fourth agent. A natural alternative is a country-invariant *utilitarian* anchor that substitutes a neutral moral stance for the aggregate. To confirm that this choice is not load-bearing, we ran a head-to-head comparison of the two variants on the twenty-country Phi-4 grid.

**Result.** The utilitarian-anchor variant improves macro MIS from 0.4252 to 0.4049 ( $\Delta = -0.0203$ ), a marginal gain that does not change any qualitative conclusion of the paper.

**Decision.** We retain the aggregate variant as the default to preserve strict country-grounding of every persona and to avoid importing a globally fixed moral prior into country-conditioned steering. The utilitarian variant remains available as an opt-in for practitioners who prefer a neutral anchor (`build_country_personas(..., fourth='utilitarian')`).

#### A16 AMCE Estimation Details

**Uniform treatment of all dimensions.** All six dimensions-Species, Gender, Age, Fitness, Social Value, and Utilitarianism-are computed identically. For each dimension, we compute the empirical mean of  $p_{\text{spare}}(\mathbf{x})$  across all scenarios in that dimension (Eq. ??). For the five binary dimensions, each scenario presents a forced choice between two character groups differing on exactly one attribute. For Utilitarianism, scenarios vary the number of lives on each side; the binary indicator is whether the larger group (“More”) or the smaller group (“Less”) is spared. This uniform mean-based estimation is consistent with the `compute_ACME` method in the MultiTP codebase [Jin et al., 2025], which fits a no-intercept linear regression with a binary group indicator-mathematically equivalent to taking the mean of the saving probability for the preferred group.

**Human AMCE scale.** The MultiTP ground-truth AMCEs in the MultiTP-released country-specific AMCE table is on a  $[-1, 1]$  scale. We convert to  $[0, 100]$  via  $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$ , where 50% represents no preference and 100% represents maximal preference for the “preferred” group. Model AMCEs are computed on the same scale.

1084 **Consistency.** Awad et al. [2018] show that marginal averages are unbiased estimators under the  
 1085 balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

## 1086 A17 Dataset Preprocessing and Sensitivity

1087 **Pipeline.** Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical  
 1088 paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where  
 1089 the two groups have identical size *and* all characters belong to the *quality-attribute* role set  
 1090 {PREGNANT, WOMAN, LARGEWOMAN}, which would otherwise inject a non-numerosity signal  
 1091 into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category  
 1092 capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled  
 1093 with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed  
 1094 seed (42).

1095 **Reproducible left/right assignment.** When the MultiTP `paraphrase_choice` field does not  
 1096 unambiguously determine which group is rendered on the left vs. right, we deterministically fall back  
 1097 to a SHA-256 hash of the tuple  $(sub_1, sub_2, g_1, g_2)$  modulo 2, so the same scenario receives the same  
 1098 ordering across runs and machines. The fallback rate is logged per country and stays below 5% on  
 1099 every country in the 20-country panel.

1100 Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing  
 1101 variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing.  
 1102 To evaluate whether these choices alter the effective evaluation distribution, we compare *six*  
 1103 preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM,  
 1104 GBR, KOR, RUS, NGA) using Llama-3.1-70B.

Table 27: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and  $< 0.010$  in 9/10 countries; the only exception is RUS at 0.013, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

| Country        | D0-Default    | D1-NoAug | D2-Cap40 | D3-Cap120 | D4-NoFlip | D5-Strict |
|----------------|---------------|----------|----------|-----------|-----------|-----------|
| USA            | 0.0655        | 0.0674   | 0.0625   | 0.0644    | 0.0643    | 0.0684    |
| DEU            | 0.0465        | 0.0515   | 0.0492   | 0.0467    | 0.0444    | 0.0484    |
| CHN            | 0.0393        | 0.0363   | 0.0385   | 0.0383    | 0.0409    | 0.0398    |
| JPN            | 0.0304        | 0.0291   | 0.0319   | 0.0296    | 0.0273    | 0.0320    |
| BRA            | 0.0492        | 0.0551   | 0.0479   | 0.0510    | 0.0520    | 0.0480    |
| VNM            | 0.0592        | 0.0574   | 0.0631   | 0.0585    | 0.0578    | 0.0626    |
| GBR            | 0.0614        | 0.0667   | 0.0608   | 0.0618    | 0.0597    | 0.0633    |
| KOR            | 0.0371        | 0.0354   | 0.0378   | 0.0357    | 0.0349    | 0.0382    |
| RUS            | 0.0383        | 0.0489   | 0.0389   | 0.0374    | 0.0383    | 0.0357    |
| NGA            | 0.0522        | 0.0568   | 0.0506   | 0.0525    | 0.0514    | 0.0538    |
| <b>Mean</b>    | <b>0.0479</b> | 0.0505   | 0.0481   | 0.0476    | 0.0471    | 0.0490    |
| $\Delta$ vs D0 | -             | +0.0025  | +0.0002  | -0.0003   | -0.0008   | +0.0011   |

1105 D1-NoAug shows the largest degradation (+0.0025), justifying synthetic augmentation of the under-  
 1106 represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating  
 1107 that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because  
 1108 it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the  $\pm 0.004$   
 1109 bootstrap CI, so reported results are not confounded by preprocessing choices.

## 1110 A18 Cross-Lingual Token Elicitation

1111 **Native-language prompting system.** Each scenario is rendered entirely in the target country’s  
 1112 native language. The implementation includes full translations of: (i) scenario framing and context  
 1113 sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms  
 1114 in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions  
 1115 with language-appropriate conjunctions and grammar.

1116 The prompt frame wraps the scenario in a native-language instruction that explicitly requests an  
 1117 English answer token. Each frame follows the pattern: “[moral dilemma preamble in native language]

1118 \n {scenario} \n [instruction to answer A or B in English]”. This ensures moral reasoning occurs in  
 1119 the culturally native linguistic frame while maintaining a consistent decision interface.

1120 **Token verification.** Token IDs are verified per model at initialisation by encoding the upper-  
 1121 case strings "A" and "B" with `add_special_tokens=False` and asserting single-token mappings:  
 1122 Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are  
 1123 extracted from the last-position output tensor; no generation sampling is performed. The model uses  
 1124 its built-in `chat_template` for correct role formatting across model families.

## 1125 A19 Baseline Implementation Details

### 1126 A19.1 Activation Steering

1127 For each country, a steering vector is computed as the difference in mean residual-stream activations  
 1128 from 20 culturally contrastive prompt pairs (e.g., “Answer as someone from [country] who values  
 1129 [WVS-high trait]” vs. “Answer as someone with the opposite values”). Vectors are extracted from  
 1130 layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding  $\alpha \cdot \mathbf{v}_{\text{steer}}$   
 1131 to the residual stream, with  $\alpha = 1.5$  selected via the same synthetic validation set used for DISCA.

1132 **Sensitivity analysis and fairness considerations.** Activation steering performance is known to be  
 1133 sensitive to (i) the extraction layer, (ii) the scaling coefficient  $\alpha$ , and (iii) the construction of contrastive  
 1134 pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers 24, 32,  
 1135 and 40 with  $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ , we found that layer 32 with  $\alpha = 1.5$  yielded the lowest  
 1136 mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We note that a more exhaustive  
 1137 search-varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-  
 1138 based direction refinement-could improve activation steering performance. However, even the  
 1139 best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch  
 1140 between linear activation interventions and the multi-dimensional moral calibration problem. The key  
 1141 structural advantage of DISCA is that it operates *per-scenario* with adaptive correction magnitude,  
 1142 whereas activation steering applies a fixed direction uniformly across all scenarios.

### 1143 A19.2 ARGS and controlled decoding (relation to our work)

1144 ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] typically assume trained  
 1145 reward models; a cross-cultural deployment would require a separate reward per target country. On  
 1146 binary forced-choice MultiTP, one can instead define rewards as closed-form functionals of persona  
 1147 logit gaps; a Prospect-Theory kernel aligned with Eq. 5 is then the natural analogue of importance-  
 1148 weighted PT-IS. We do not run the full ARGS decoding stack in our experiments; Table 17 and §6  
 1149 isolate the kernel comparison on the shared DISCA pipeline.

## 1150 A20 Trigger Mechanism and Latency

1151 DISCA runs dual-pass IS on every scenario; inter-persona variance is logged but does not gate  
 1152 sampling. The *flip rate* (IS reversals vs. consensus) is 29.0% in the cross-backbone ablation suite  
 1153 (Table 5), indicating confidence modulation rather than wholesale decision reversal.

1154 A compact round-2 latency benchmark on H100 shows similar overhead across proposal budgets:  
 1155 0.086s/scenario at  $K=64$ , 0.083s at  $K=128$ , and 0.083s at  $K=256$ , compared with 0.023s for vanilla  
 1156 decoding ( $3.57\text{--}3.72\times$  overhead).

Table 28: Inference latency benchmark (Phi-4, H100).

| Method   | $K$ | sec/scenario | Overhead     |
|----------|-----|--------------|--------------|
| swa_dpbr | 64  | 0.086        | $3.72\times$ |
| swa_dpbr | 128 | 0.083        | $3.59\times$ |
| swa_dpbr | 256 | 0.083        | $3.57\times$ |
| vanilla  | —   | 0.023        | $1.00\times$ |

1157 **Cost-vs-quality frontier across backbones.** Figure 5 plots the *deployed-cost* side of the Phi-4-  
1158 vs-Llama narrative: at 350 ms per scenario, Phi-4 with DISCA reaches MIS = 0.346, while Llama-  
1159 3.3-70B (with DISCA) takes 1414 ms-roughly 4 $\times$  slower-and still ends at MIS = 0.668. Phi-4 thus  
1160 Pareto-dominates the 70B baseline on *both* axes simultaneously: better alignment at lower latency.  
1161 Among the headline seven, only Magistral-24B sits on a comparable quality plateau, at almost double  
1162 Phi-4’s per-scenario cost.

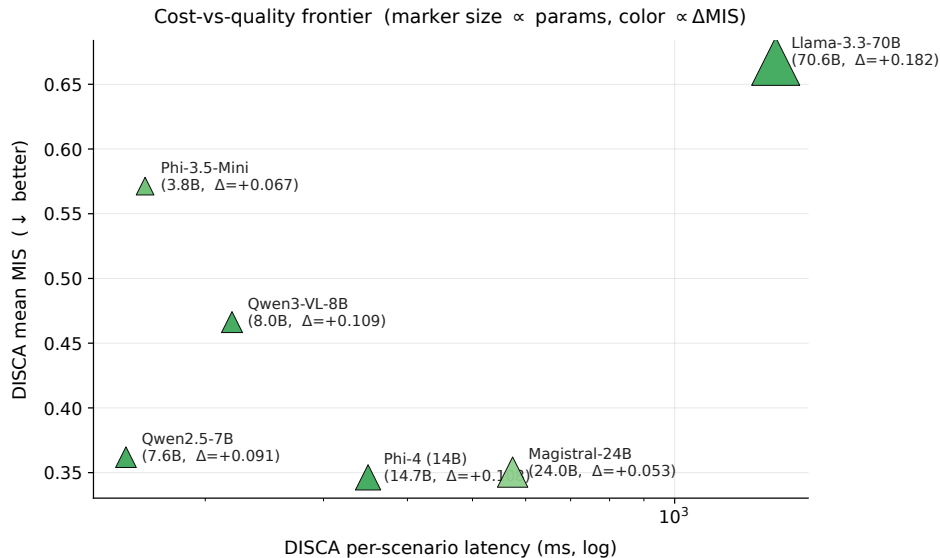


Figure 5: **Cost-vs-quality frontier on the headline 7 models.** Per- scenario DISCA latency (log scale) vs. mean DISCA MIS. Marker size is proportional to parameter count; color encodes  $\Delta$ MIS (greener = larger DISCA gain). Phi-4 (14B,  $\Delta = +0.108$ ) lies bottom-left: Pareto-dominant over Llama-3.3-70B in both latency and alignment.

## 1163 A21 Engineering Journey: Method Selection

1164 DISCA (the “EXP-24” configuration) was selected from **25 method variants** developed and evaluated  
1165 on a fixed three-model, five-country prototyping panel (Qwen2.5-7B, Gemma-2-9B, Mistral-7B  $\times$   
1166 USA, CHN, JPN, DEU, BRA). Table 29 summarises the top-performing variants.

Table 29: Method variant comparison on the 3-model  $\times$  5-country prototyping panel (15 rows). Mean MIS  $\downarrow$ . All variants share the same persona construction and debiasing; they differ in IS-stage design.

| Rank | Method variant                            | Mean MIS $\downarrow$ | Key innovation                      |
|------|-------------------------------------------|-----------------------|-------------------------------------|
| 1    | EXP-23: Category-Coherence Reg.           | 0.3956                | Blend IS toward per-cat mean        |
| 2    | <b>EXP-24: Dual-Pass Bootstrap (DPBR)</b> | <b>0.3969</b>         | Soft $r = \exp(-v_b/s)$ reliability |
| 3    | EXP-09: Hierarchical IS                   | 0.3975                | Cross-scenario smoothing (legacy)   |
| 4    | EXP-22: Adaptive IS $\sigma$              | 0.3977                | $\sigma$ from agent/history stats   |
| 5    | EXP-16: Nesterov IS                       | 0.3977                | Momentum + progressive PT           |
| 6    | EXP-17: Dual-Momentum Prior               | 0.3979                | Fast/slow $\beta$ EMA blend         |
| 7    | EXP-10: Grand Fusion                      | 0.3982                | EXP-03+05+09 combined               |
| 18   | EXP-01: Base SWA-PTIS                     | 0.4269                | Single-pass IS baseline             |
| 25   | EXP-12: Contrastive Personas              | 0.4517                | World-average subtraction           |

1167 The top cluster (ranks 1–7) is separated by only 0.0026 MIS, but DPBR was selected over the  
1168 marginally-better EXP-23 for two reasons: (i) DPBR materially **lowers flip rate** (the fraction of  
1169 scenarios where IS reverses the consensus decision) compared to all other top-cluster methods, im-  
1170 proving deployment stability; and (ii) the dual-pass bootstrap provides an **interpretable uncertainty**  
1171 **signal** ( $r$  in Eq. 6) that can be logged and audited per scenario. The base SWA-PTIS (rank 18) shows  
1172 the value added by the dual-pass stabilization stages: a 7% relative MIS improvement. Contrastive

1173 persona decoding (rank 25) *hurts* alignment, validating the design choice discussed in §2 to use direct  
1174 WVS personas rather than contrastive amplification.

## 1175 A22 Relationship to Persona-Dependent LLM Alignment

1176 ? is the closest prior work to DISCA in studying persona effects on LLM moral decisions. Their study  
1177 and ours share the Moral Machine framework and AMCE-based evaluation, but differ fundamentally  
1178 in goal, method, and scope. Table 30 summarises the key distinctions.

Table 30: Comparison between ? and DISCA.

| Dimension      | Kim et al. (2025)               | DISCA (this work)                  |
|----------------|---------------------------------|------------------------------------|
| Goal           | Measure persona sensitivity     | Correct cultural misalignment      |
| Personas       | 7 sociodemographic categories   | 4 WVS-grounded agents per country  |
| Persona source | Author-designed binary labels   | Empirical WVS-7 microdata          |
| Metric         | MDD (persona-pair distance)     | MIS ( $\ell_2$ to human AMCE)      |
| Correction     | None (diagnostic only)          | PT-IS + dual-pass reliability gate |
| Models         | 3 (GPT-4o, GPT-3.5, Llama-2)    | 7 checkpoints, 5 families, 2B–70B  |
| Countries      | Aggregate (Western vs. Eastern) | 20 individual countries            |
| Scenarios      | 9 dimensions, 10k synthetic     | 6 dimensions, 310–500 per country  |
| Theory         | Partisan sorting (descriptive)  | James–Stein shrinkage (Thm. 1)     |

1179 Three findings from ? directly inform DISCA’s design.

1180 **Persona sensitivity motivates disagreement-driven steering.** Kim et al. show that LLMs exhibit  
1181 moral-decision distances (MDD)  $2\text{--}4\times$  larger than humans under contrasting personas. This confirms  
1182 that persona prompts do shift LLM moral reasoning substantially, validating the premise that within-  
1183 country persona spread is an informative signal. DISCA converts this sensitivity from a liability  
1184 (uncontrolled decision shifts) into a feature (a sufficient statistic for correction reliability, Theorem 1).

1185 **The partisan sorting phenomenon justifies WVS grounding.** Kim et al. find that political  
1186 orientation produces the largest persona effect in LLMs, far exceeding its effect in human respondents.  
1187 This “partisan sorting” effect reveals that LLMs over-index on politically salient dimensions when  
1188 given sociodemographic labels. DISCA avoids this pathology by grounding personas in empirical  
1189 WVS-7 microdata rather than author-designed binary labels: the ten-dimensional cultural profile  
1190 (Table 24) distributes influence across religiosity, child-rearing values, social trust, and other axes,  
1191 preventing any single dimension from dominating the correction. The leave-one-WVS-dim-out probe  
1192 (Table 26) confirms that influence is distributed rather than concentrated.

1193 **Moral flips validate the reliability gate.** Kim et al. document that specific personas can entirely  
1194 reverse LLM preferences on certain moral dimensions (e.g., social status under a progressive persona  
1195 in GPT-4o). DISCA’s dual-pass reliability gate (Eq. 6) is designed precisely for this regime: when  
1196 persona-driven corrections are large enough to flip the decision, the two independent IS passes  
1197 are likely to disagree, and the exponential decay weight  $r$  shrinks the correction toward zero. The  
1198 tail-safety analysis (Table 13) confirms this mechanism reduces harmed cells from 11/120 to 3/120  
1199 compared to ungated consensus.

1200 **From diagnosis to correction.** The fundamental advance of DISCA over the Kim et al. framework  
1201 is operational: where they diagnose persona sensitivity, we harness it. Their MDD metric measures  
1202 the distance between two contrasting personas; our MIS measures the distance to the human target  
1203 and actively reduces it. The James–Stein result (Theorem 1) provides the formal bridge: within-panel  
1204 disagreement (analogous to their MDD) is a sufficient statistic for the correction’s reliability, and the  
1205 shrinkage estimator converts this diagnostic signal into an optimal correction.

## 1206 **A23 Extended Limitations**

1207 This appendix expands the limitations summarised in §6 with technical caveats not covered in the  
1208 main text.

1209 **WVS-to-trolley linkage.** The mapping from WVS value dimensions to trolley moral dimensions  
1210 operates through both direct thematic overlap and indirect modulation (Appendix A15.3). The  
1211 10-dimensional WVS profile explains  $R^2 \in [0.55, 0.69]$  of human AMCE variance, and a leave-  
1212 one-dimension-out probe identifies load-bearing couplings alongside noisy ones that the IS stage  
1213 automatically down-weights. We avoid claims of the form “WVS dimension  $X$  caused AMCE shift  
1214  $Y$ ,” but the ensemble design and the IS noise-filtering mechanism together ensure the method is  
1215 robust even when individual WVS–trolley links are imperfect.

1216 **Inter-persona reward assumption.** The reward  $r_i = \delta_i - \delta_{\text{base}}$  assumes persona shifts approximate  
1217 human target directions. This is plausible when the base model represents a WEIRD-biased prior, but  
1218 if a persona’s shift is orthogonal to the human target, the signal may mislead. Quantifying whether  
1219 persona-consensus directions correlate with country-level human AMCE vectors independently of  
1220 the base model-and when corrections diverge from targets-is open; we do not report that correlation  
1221 matrix here.

1222 **Geographic coverage, preprocessing, and WVS vintage.** We report 20 countries out of 100+  
1223 MultiTP countries; results depend on the preprocessing pipeline in Appendix A17. WVS Wave 7  
1224 ( $\sim 2017\text{--}2022$ ) is a fixed historical slice that may not reflect current preferences in rapidly evolving  
1225 societies; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected  
1226 in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel;  
1227 cross-lingual generalisation to such settings remains untested, and one language per country may  
1228 conflate linguistic and cultural effects.

1229 **Model quantisation.** 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability-  
1230 observed empirically on Unsloth Phi-4 prior to switching to vLLM BF16 for the headline runs.